

Unrestricted Faithful Rounding is Good Enough for Some LNS Applications

Mark Arnold

Colin Walter

**University of Manchester Institute
of Science and Technology**



Outline

Why choose Logarithmic Number Systems (LNS)?

Floating Point versus LNS

Round to Nearest is Hard

Restricted versus Unrestricted Faithful Rounding

FFT simulation

Multimedia Example

Conclusions

Arithmetic Choices

Fixed-point (FX)

Scaled integer—manual rescale after multiply

Hard to design, but common choice for cost-sensitive applications

Floating-point IEEE-754 (FP)

Exponent provides automatic scaling for mantissa

Easier to use but more expensive

Logarithmic Number System (LNS)

Converts to logarithms once—keep as log during computation

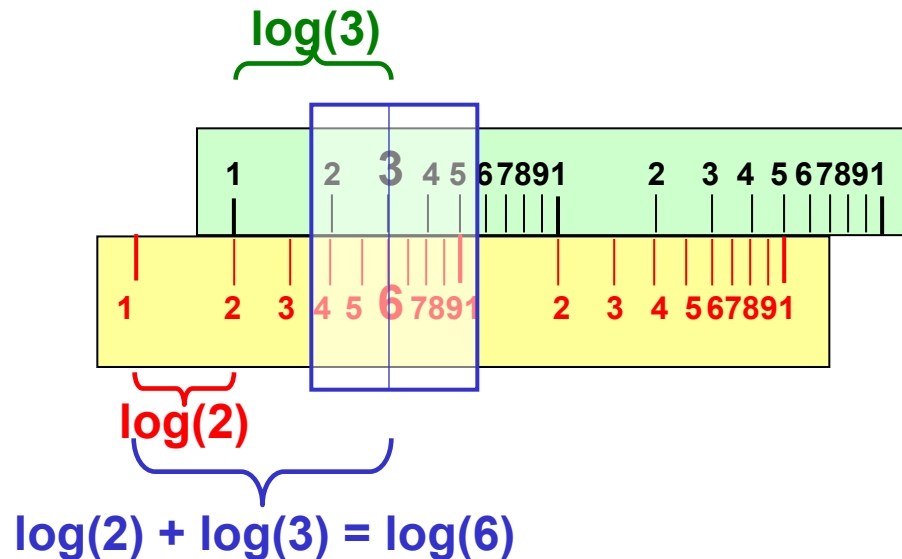
Easy as FP, can be faster, cheaper, lower power than FX

Advantages of LNS

Cheaper multiply, divide, square root

Good for applications with high proportion of multiplications

Multimedia example coming



Most significant bits change less frequently: power savings

Commercial Interest in LNS

Motorola:	120MHz LNS 1GFLOP chip [pan99]
European Union:	LNS microprocessor [col00]
Yamaha:	Music Synthesizer [kah98]
Boeing:	Aircraft controls
Interactive Machines,Inc.:	IMI-500: Animation for <i>Jay Jay the Jet Plane</i>

Notation

x = real **values**,

X = corresponding logarithmic **representations**

b = base of the logarithm ($b=2$ is typical)

F = precision

$\varepsilon = 2^F \sqrt[b]{b}$, i.e., the smallest value > 1.0

if x is exact LNS value, $x\varepsilon$ is next larger exact value

$\mu = \log_b(\varepsilon)$, the LNS representation of ε

X and $X+\mu$ are the corresponding representations

LNS Addition

Given $X = \log_b(x)$ and $Y = \log_b(y)$:

1. Let $Z = X - Y$
2. Lookup $s_b(Z) = \log_b(1 + b^Z)$
3. $T = Y + s_b(Z)$

Why it works:

1. $Z = \log_b(x/y)$
2. $s_b(Z) = \log_b(1 + x/y)$
3. $T = \log_b(y(1 + x/y))$

Thus, $T = \log_b(y + x)$

Hardware:

1 subtractor

1 function approximation unit

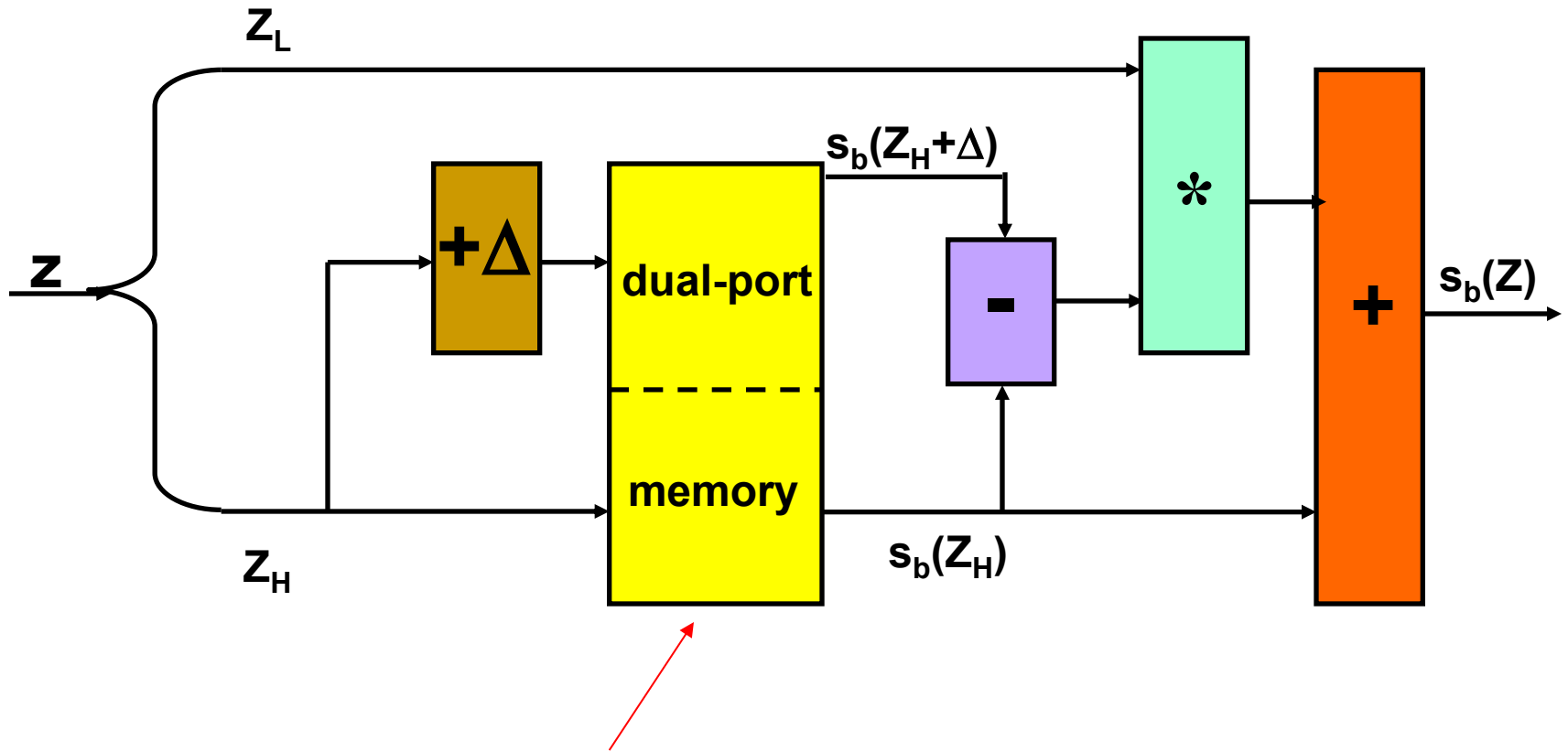
lookup table (ROM or RAM) for $F < 12$

interpolation for higher precision

1 adder

Similar function, d_b , for subtraction

Stored Value Linear Interpolation

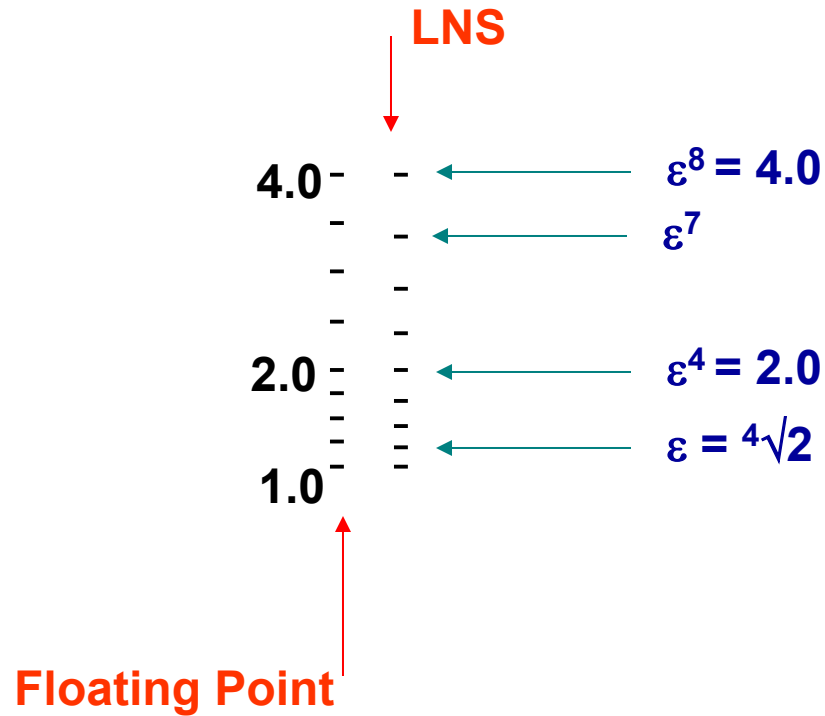


Virtex FPGA has dual-port block RAM

Can obtain adjacent tabulated points in one cycle

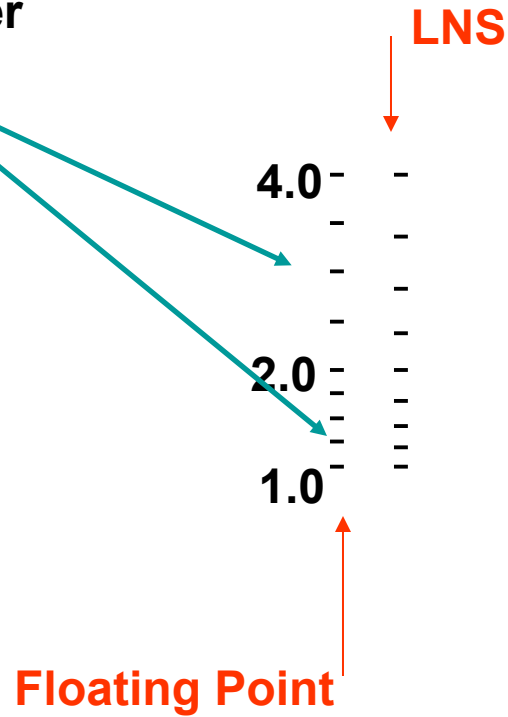
Floating Point versus LNS

Exactly representable points shown for precision $F=2$



Floating Point versus LNS

Floating point has greater relative error here



Floating Point versus LNS

Floating point has greater
relative error here

LNS

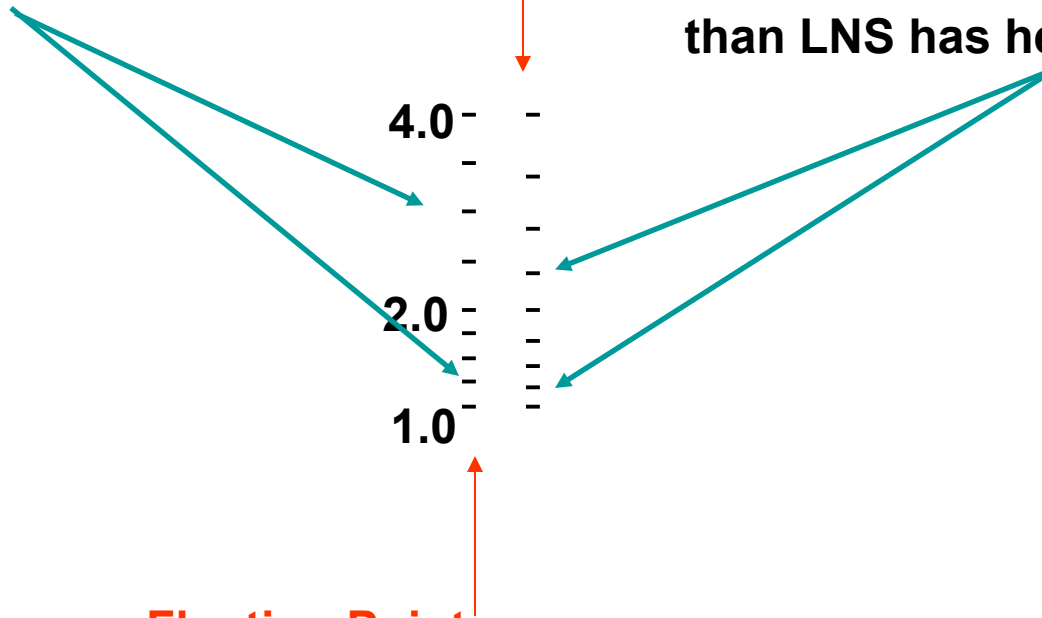
than LNS has here

4.0

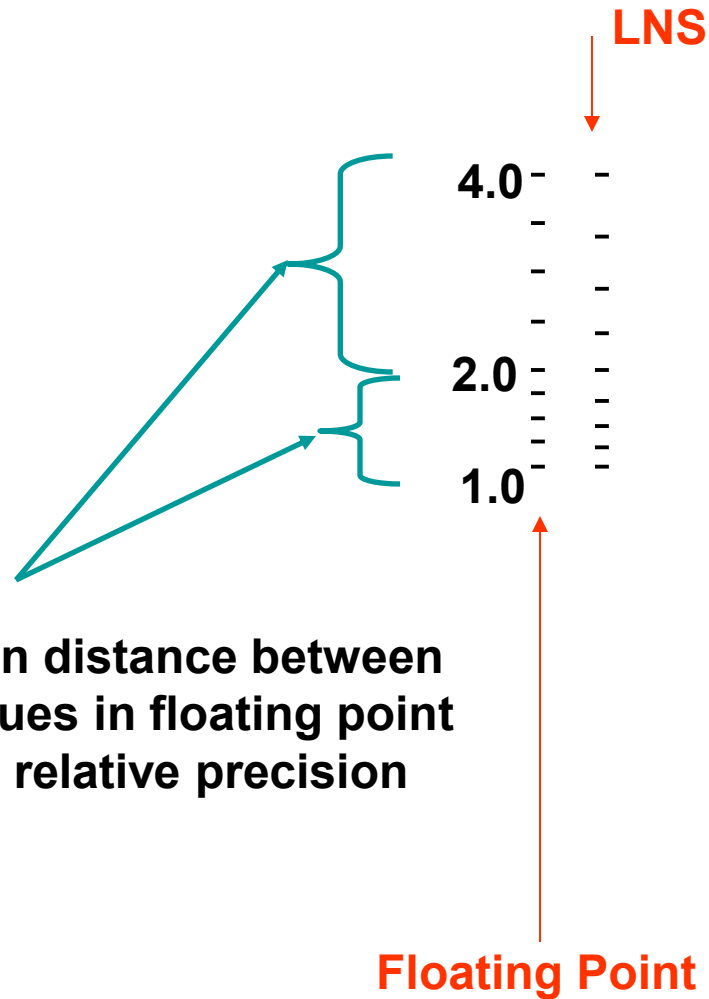
2.0

1.0

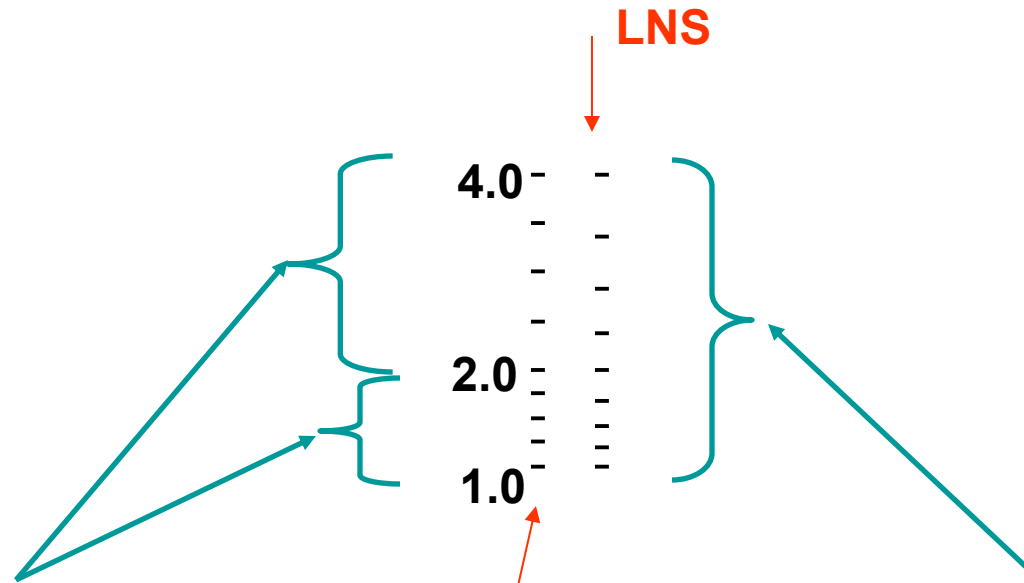
Floating Point



Floating Point versus LNS



Floating Point versus LNS

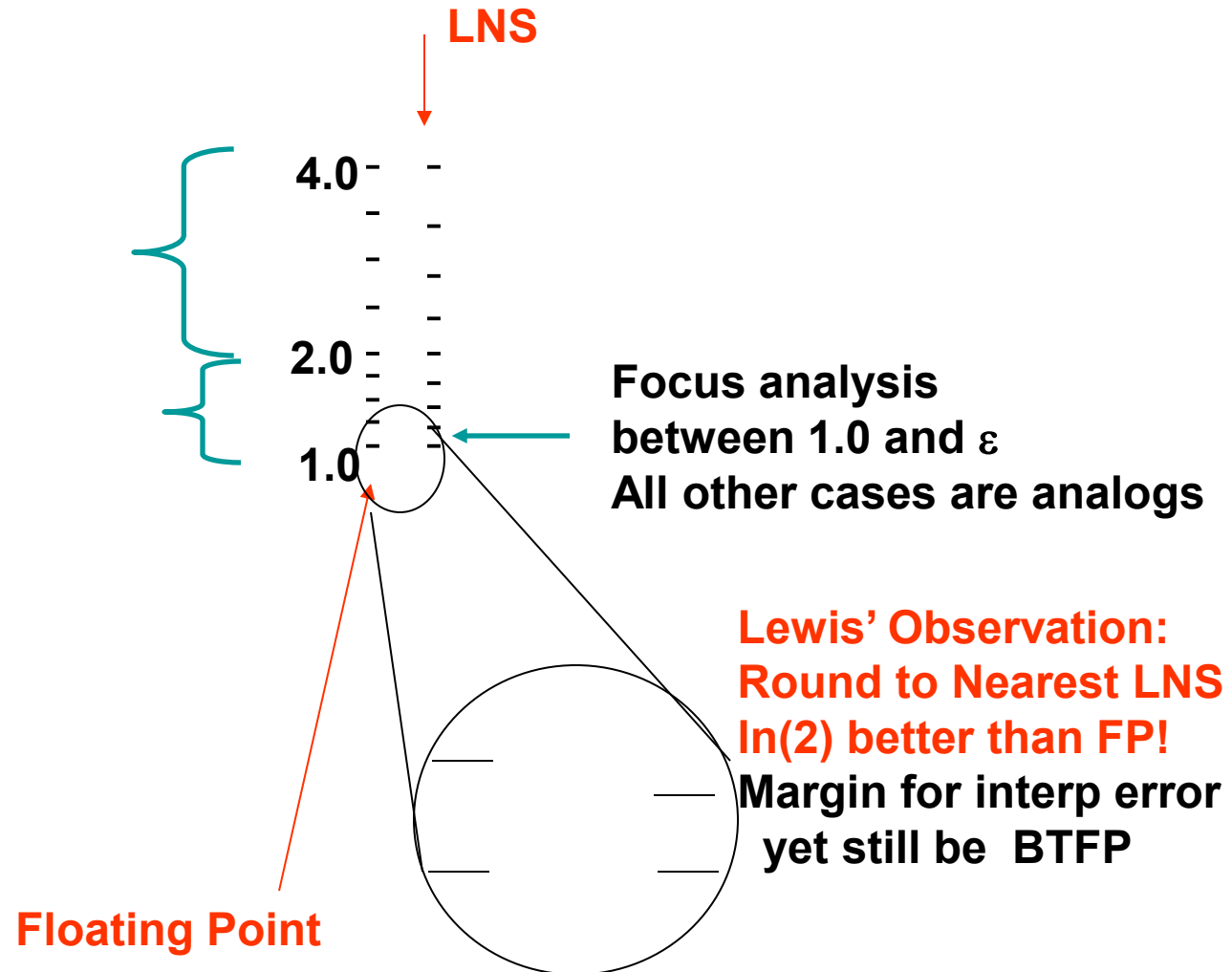


Discrete change in distance between representable values in floating point causes wobble in relative precision

Continuous change in distance between representable values in LNS means constant relative precision

Floating Point

Floating Point versus LNS



Rounding Modes

Round to Nearest

Prescribed by IEEE-754 for Floating Point (FP)

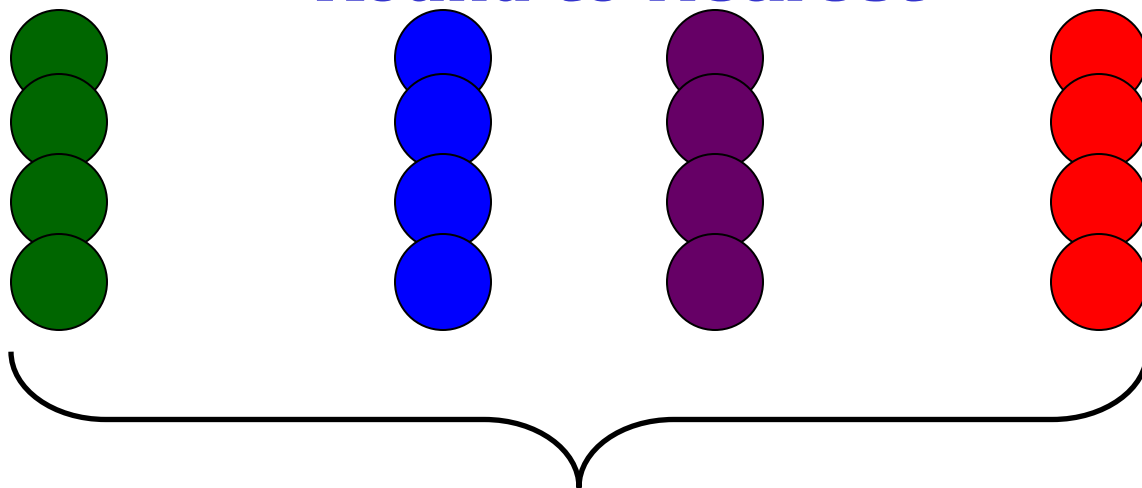
Affordable for FP at any precision

Economical for LNS only at low precision ($F < 12$)

Restricted Faithful

Unrestricted Faithful

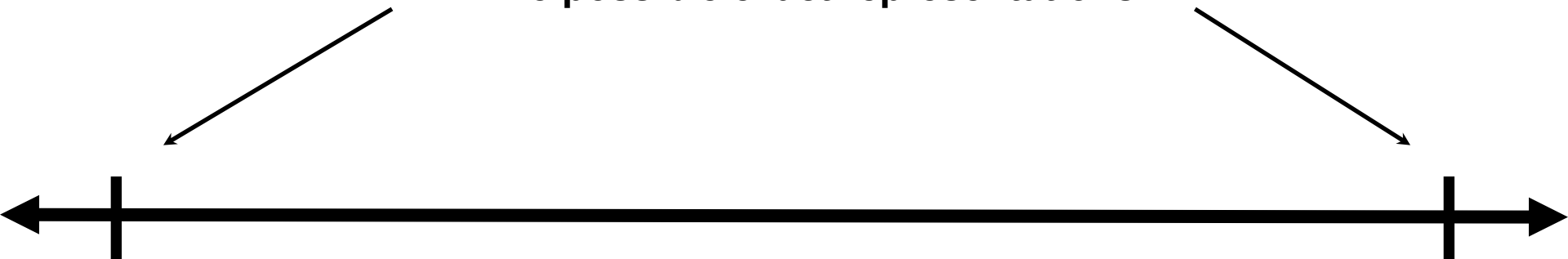
Round to Nearest



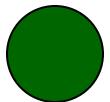
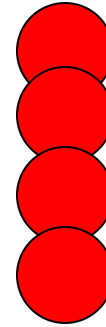
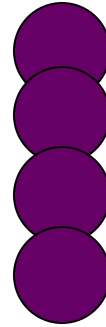
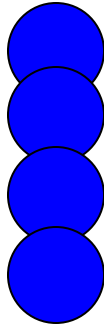
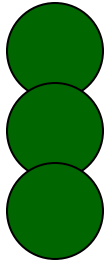
Non-exactly-representable values

round to the nearest of the

Two possible exact representations



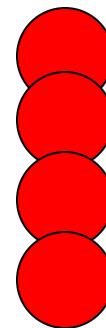
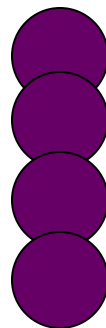
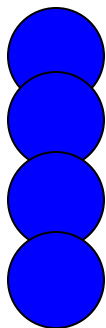
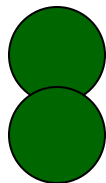
Round to Nearest



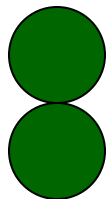
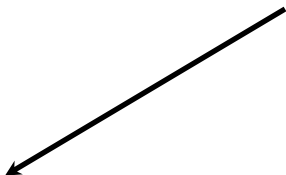
The green point is
closer to the left
representation



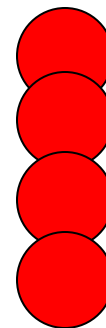
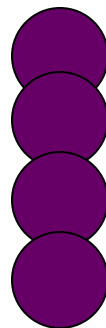
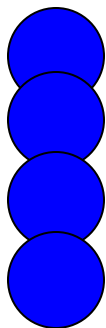
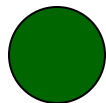
Round to Nearest



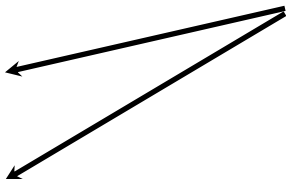
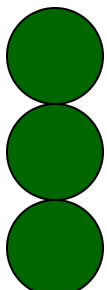
**Similar values will
round to the same
representation**



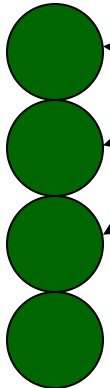
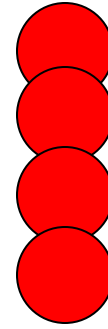
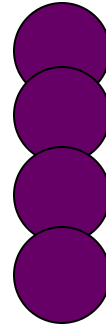
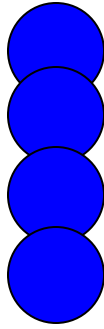
Round to Nearest



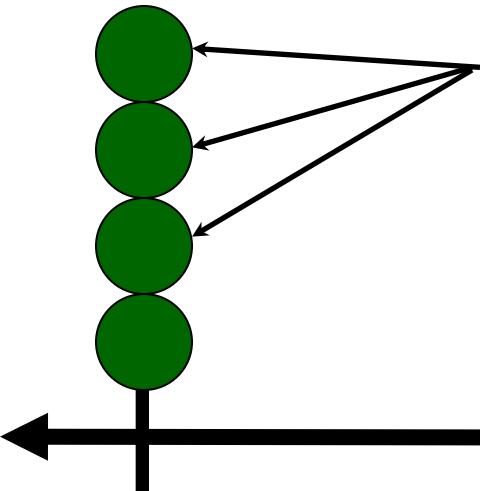
**Similar values will
round to the same
representation**



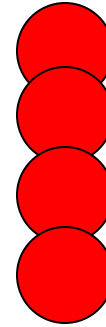
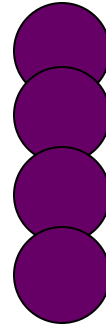
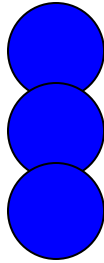
Round to Nearest



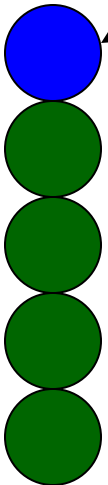
**Similar values will
round to the same
representation**



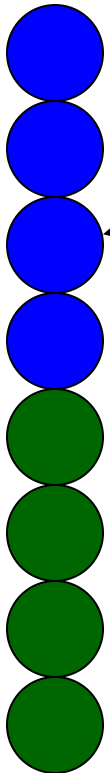
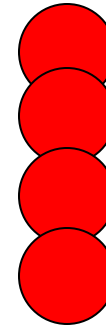
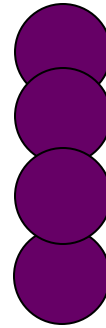
Round to Nearest



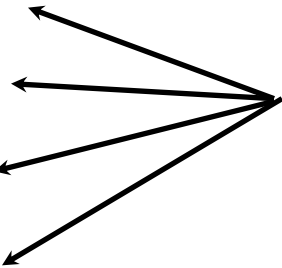
All values on the left,
no matter how close
to the midpoint, round
to this representation



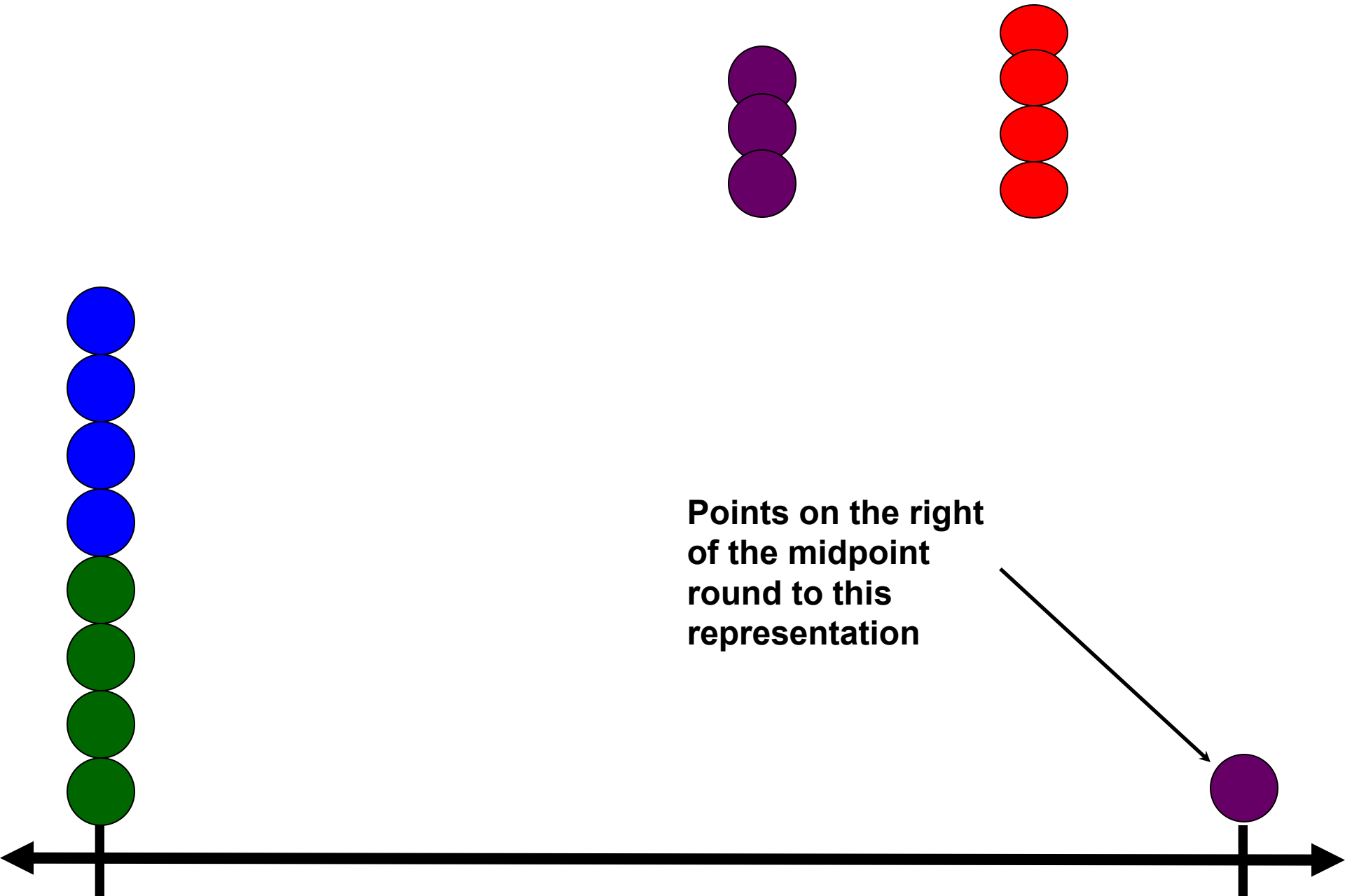
Round to Nearest



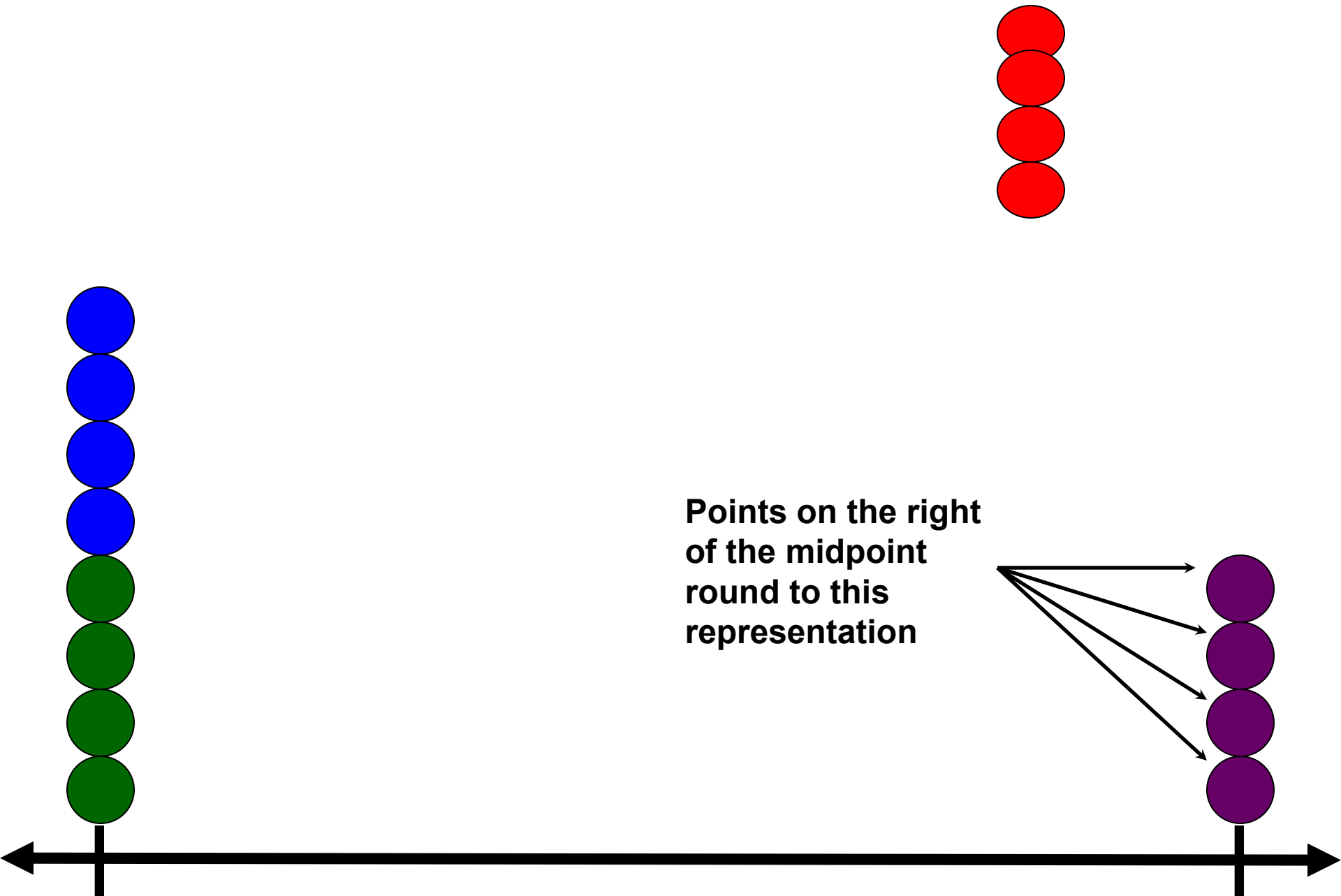
All values on the left,
no matter how close
to the midpoint, round
to this representation



Round to Nearest



Round to Nearest



Round to Nearest

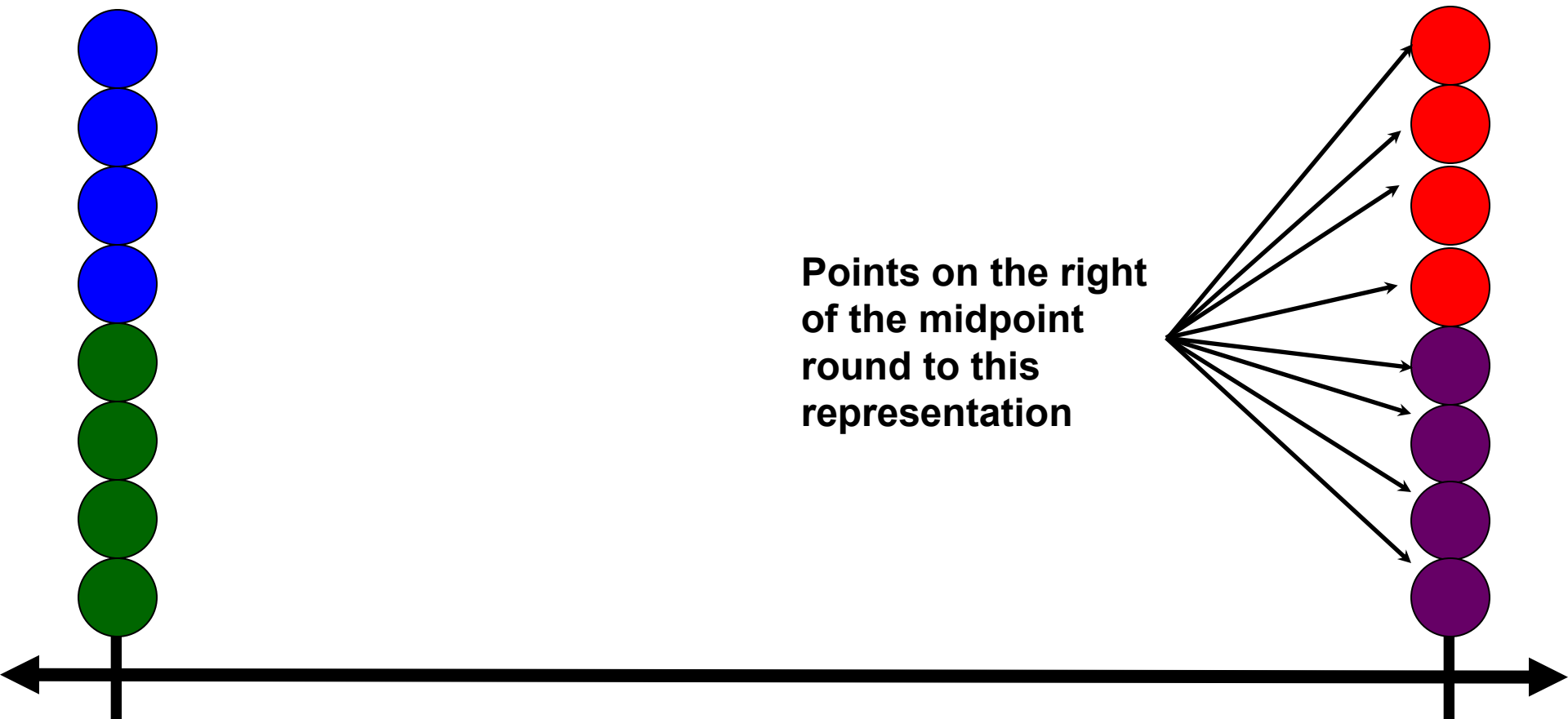


Table Makers' Dilemma

Need interpolation of s_b and d_b for moderate to high precision

some results will be hard to round to nearest

would cost much more memory

Relax rounding requirements

Faithful rounding chooses one of two closest points

More next-nearest points decreases memory requirements for s_b

Table Requirements for 32-bit Addition

Proposed	Unrestricted Faithful	234	(words)
Lewis	Restricted Faithful	768	
Coleman et al.	Restricted Faithful	≈ 1500	
Swartzlander	Round to Nearest	$\approx 2^{28}$	

Faithful Rounding Modes

Restricted Faithful

Higher LNS Precision ($F=23$) than Round-to-Nearest FP
“Better than Floating Point” (BTFP) in worse case
Like Round-to-Nearest except near midpoint

Unrestricted Faithful

Allowed by the Brown Model for Floating Point
Proposed here as “good enough” for some LNS apps
Cuts LNS memory size 3- to 6- fold vs. Restricted

Probabilistic Model

p = probability faithful result does not round to the nearest

Unrestricted Faithful

$$0 \leq p \leq 1.0$$

Restricted Faithful

$$0 \leq p \leq \rho$$

δ = distance (in log domain) from midpoint in which
rounding either way is permitted

$$\rho = \delta / (\mu / 2) = .443 = \text{max probability allowed for BTFP}$$

$p = .443$ acts like FP

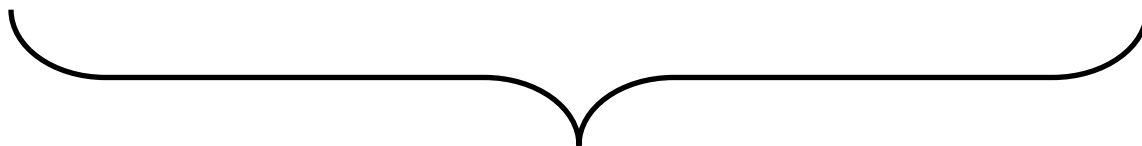
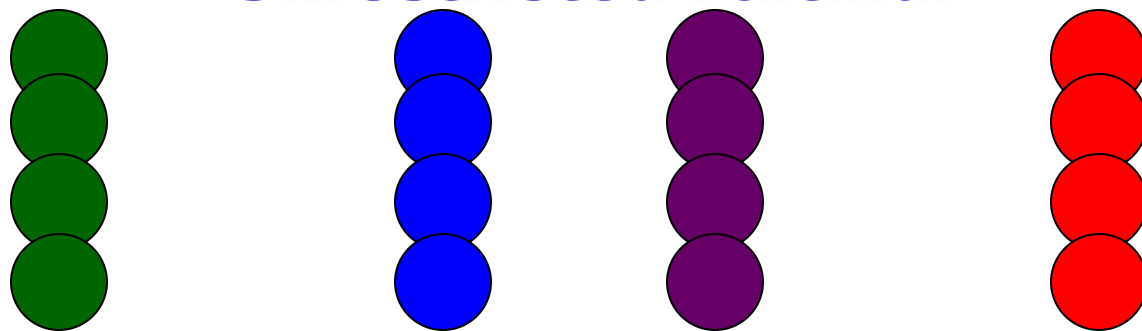
Round to Nearest

$$p = 0$$

Restricted or Unrestricted

Unrestricted Faithful

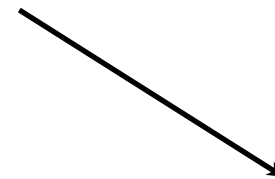
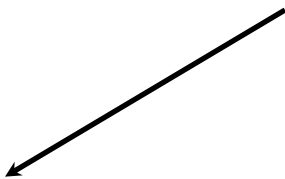
$p = .25$



Non-exactly-representable values

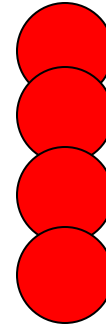
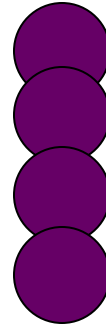
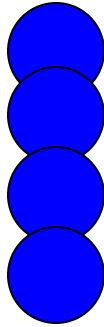
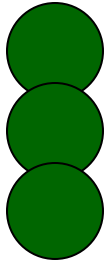
round to either of the

Two possible exact representations

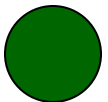


Unrestricted Faithful

$p = .25$

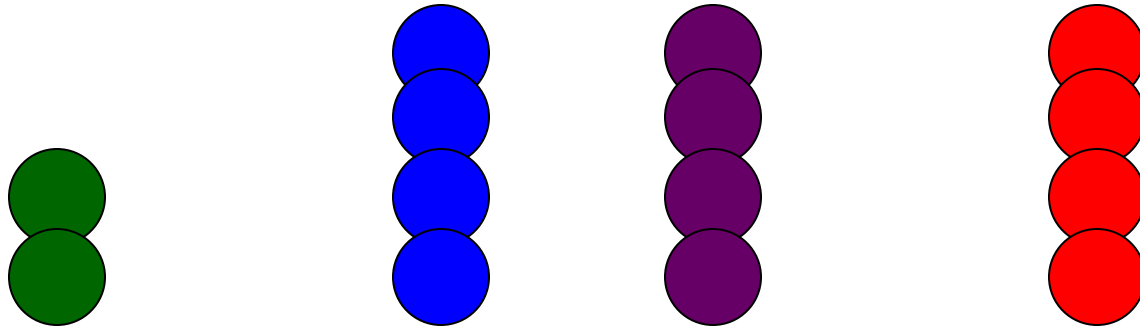


The first **green**
point rounds to
the more accurate
representation



Unrestricted Faithful

p = .25

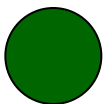
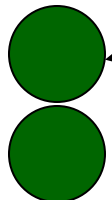
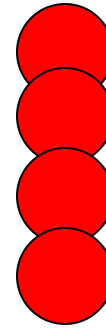
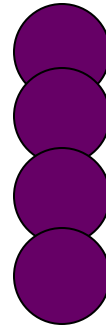
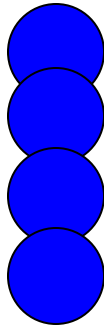
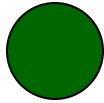


A similar **green**
value rounds to
the less accurate
representation



Unrestricted Faithful

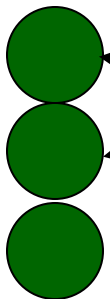
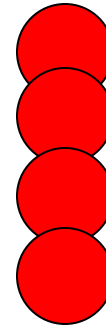
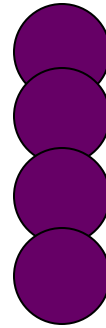
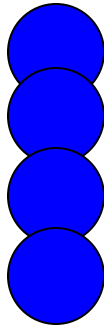
$p = .25$



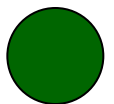
On average, 3 out
of 4 **green** points
will round to the
nearest because
 $p = .25$

Unrestricted Faithful

$p = .25$

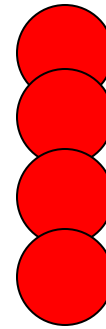
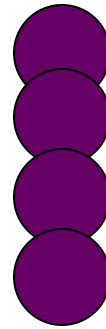
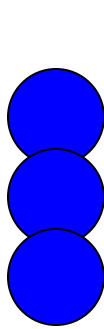


On average, 3 out
of 4 **green** points
will round to the
nearest because
 $p = .25$

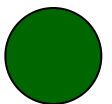
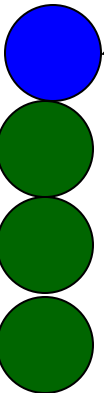


Unrestricted Faithful

$p = .25$

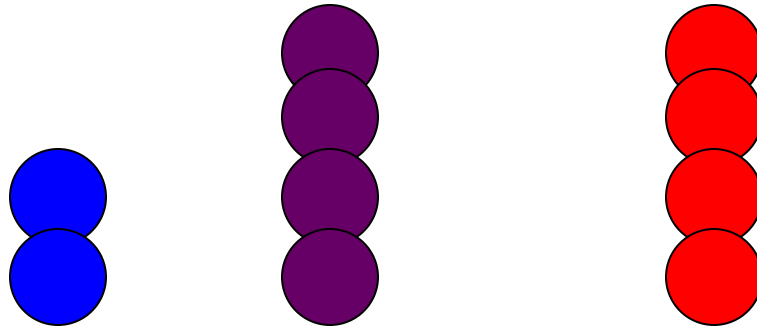


Values slightly left
of the midpoint
(blue) generally
round here

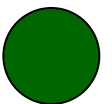
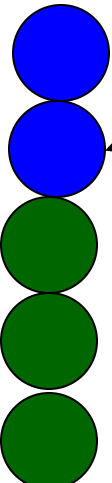


Unrestricted Faithful

$p = .25$

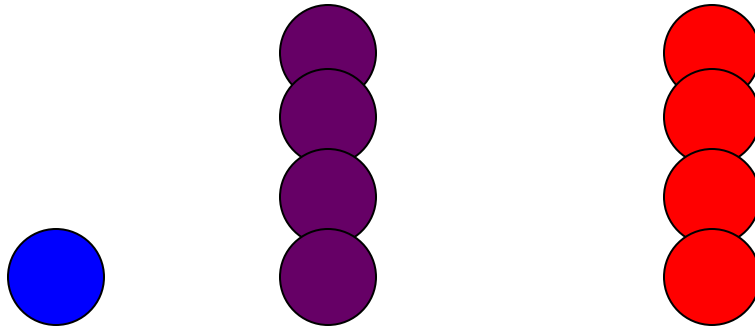


Values slightly left
of the midpoint
(blue) generally
round here



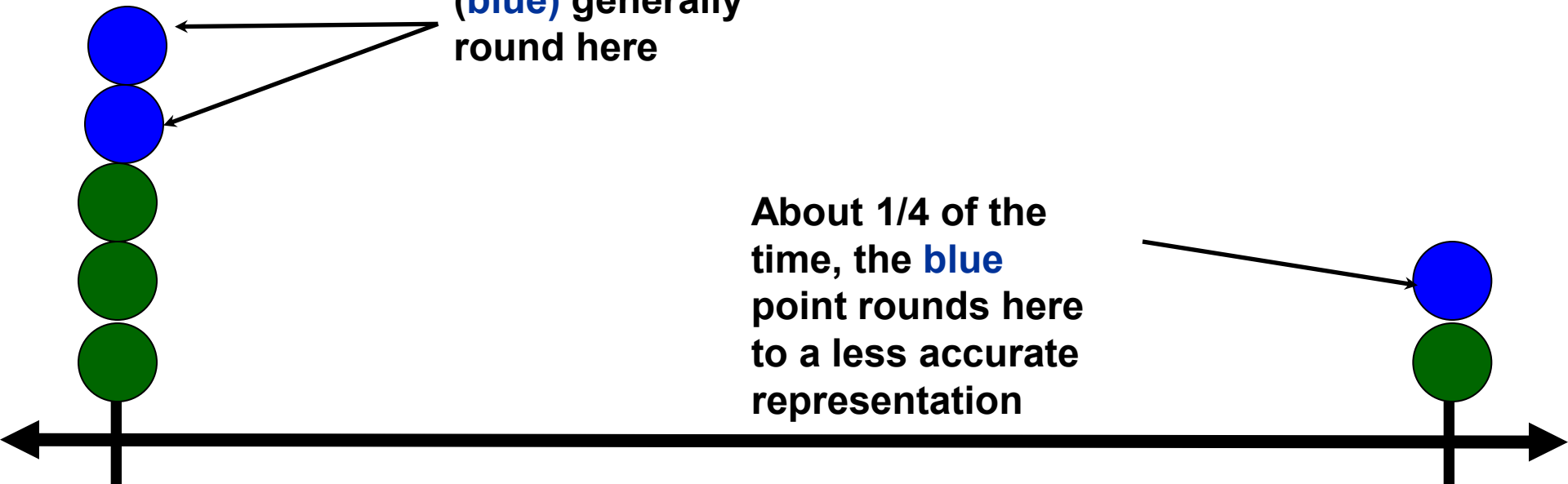
Unrestricted Faithful

$p = .25$



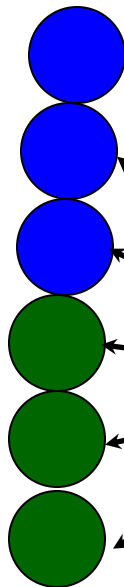
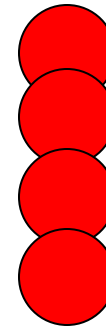
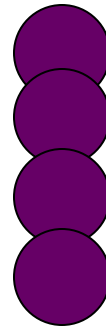
Values slightly left
of the midpoint
(blue) generally
round here

About 1/4 of the
time, the blue
point rounds here
to a less accurate
representation



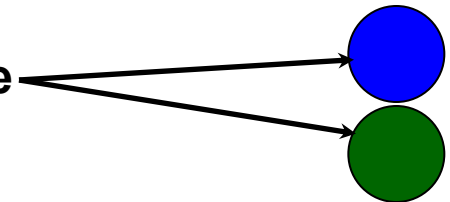
Unrestricted Faithful

$p = .25$



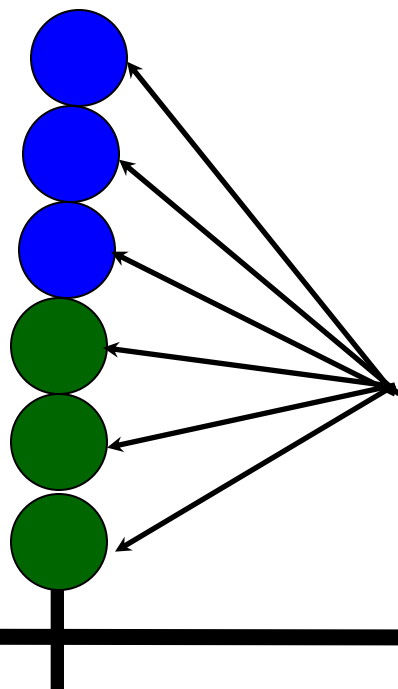
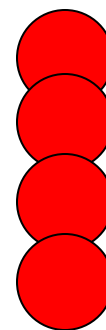
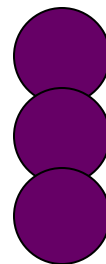
3/4 of the points to the left of the midpoint are rounded to the nearest

1/4 of the points to the left of the midpoint are rounded to the next-nearest

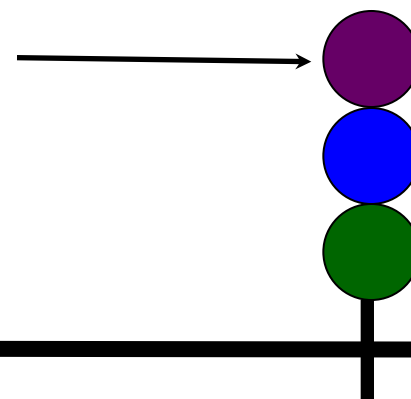


Unrestricted Faithful

p = .25

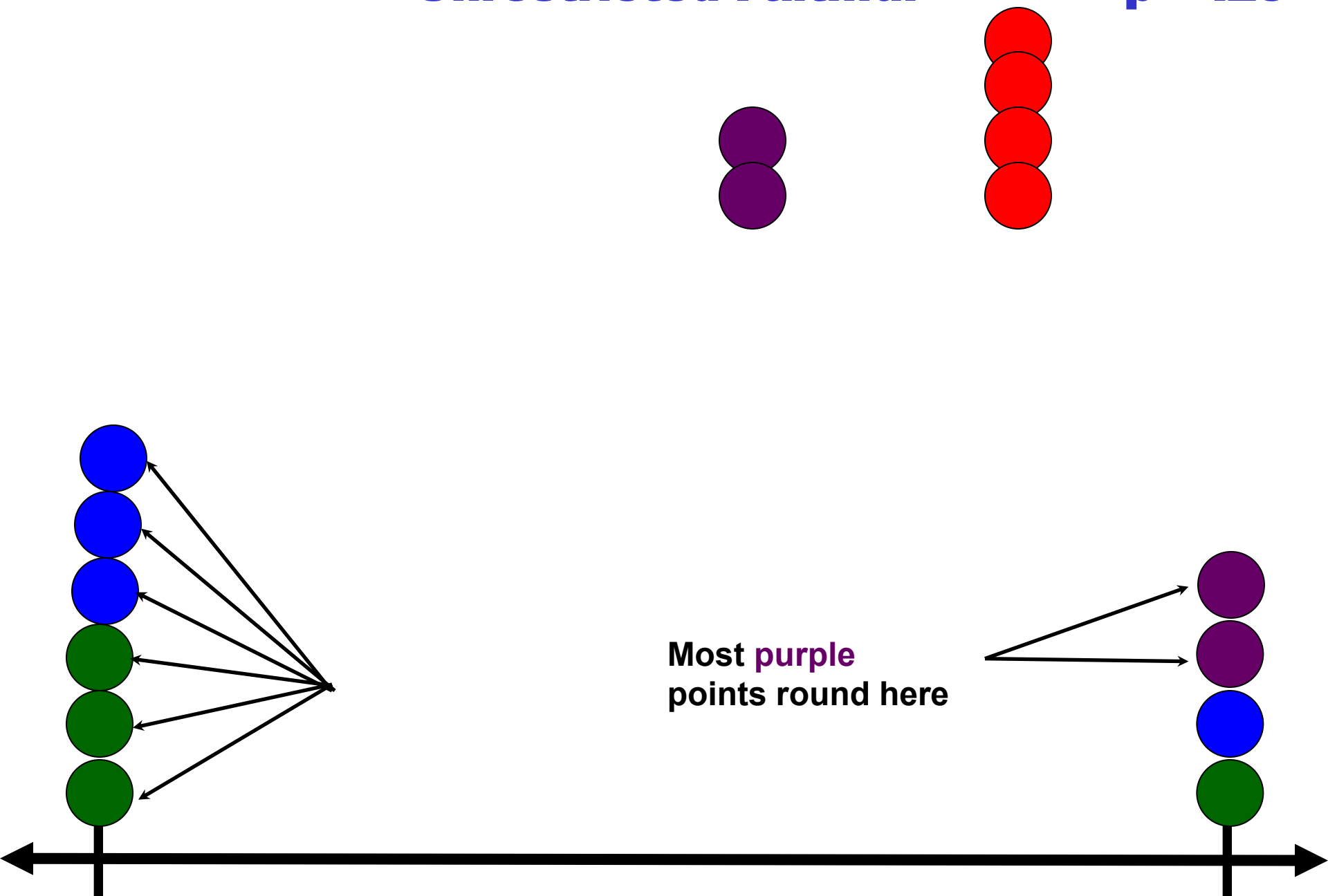


Most **purple**
points round here



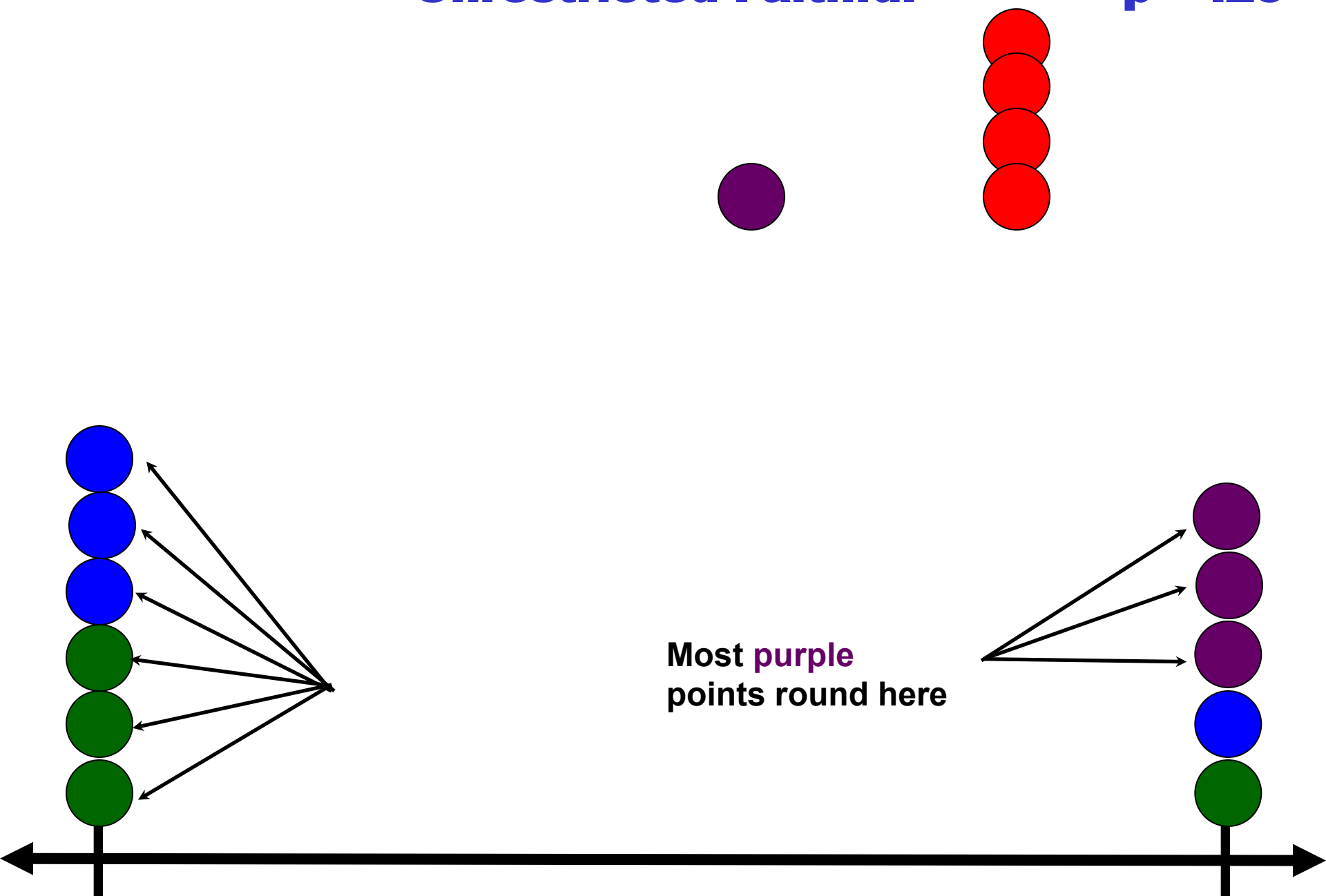
Unrestricted Faithful

p = .25



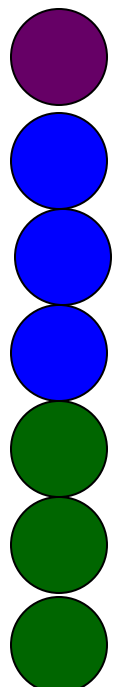
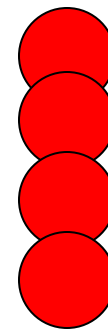
Unrestricted Faithful

p = .25



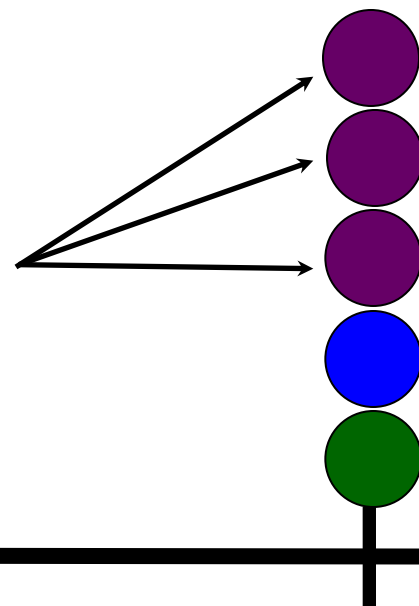
Unrestricted Faithful

$p = .25$



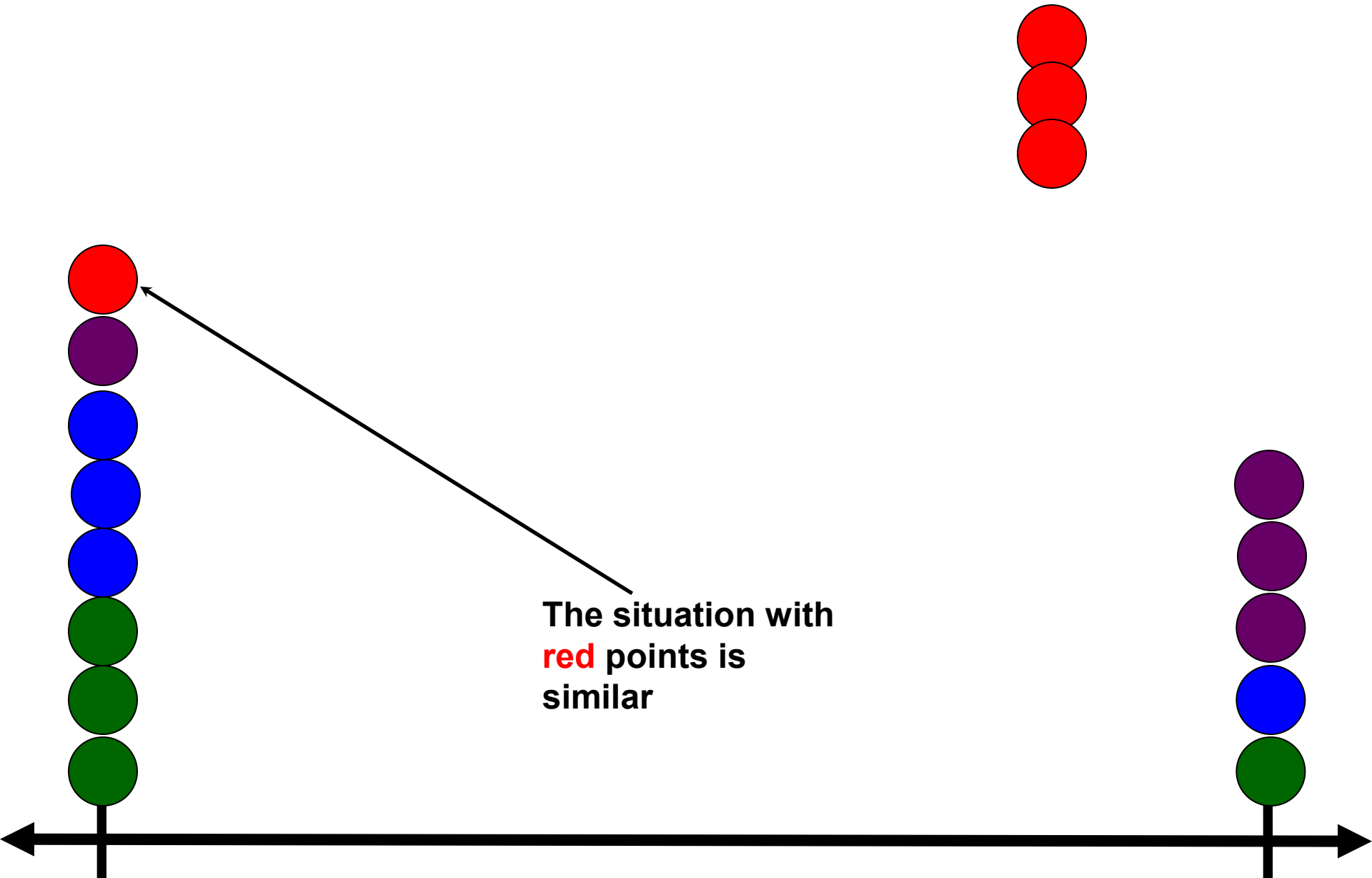
An occasional
purple point
rounds to the next
nearest
representation

Most **purple**
points round here



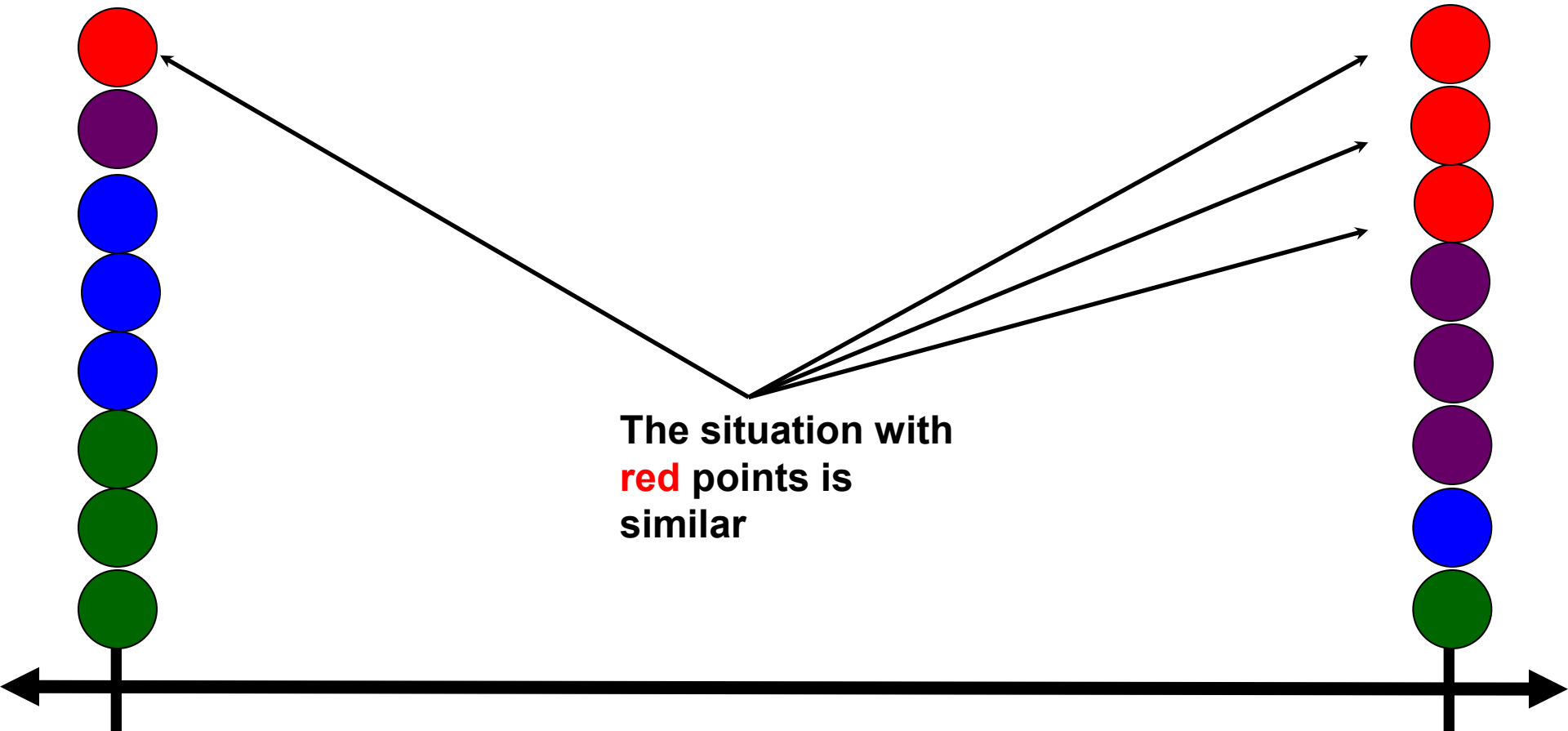
Unrestricted Faithful

$p = .25$



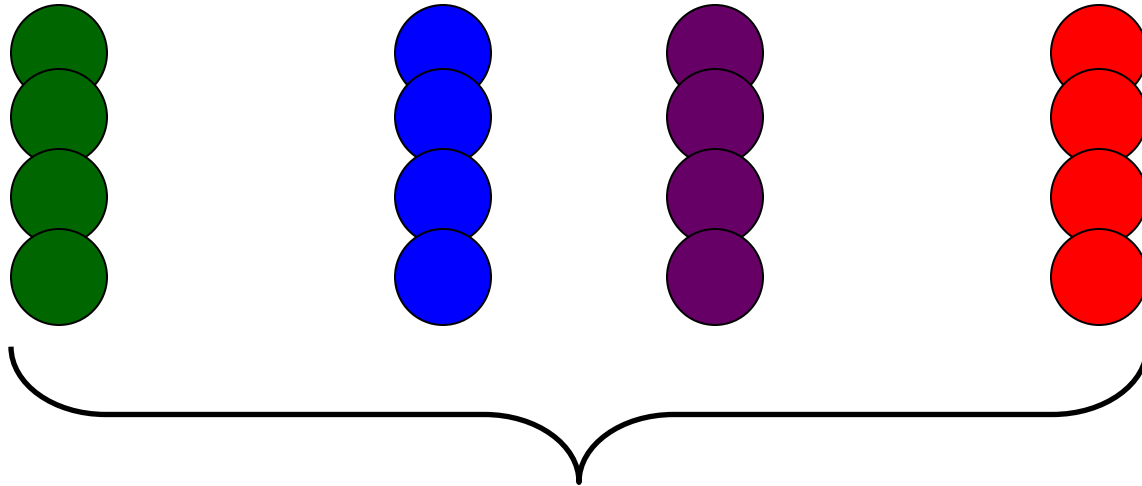
Unrestricted Faithful

$p = .25$



Restricted Faithful

p = .25

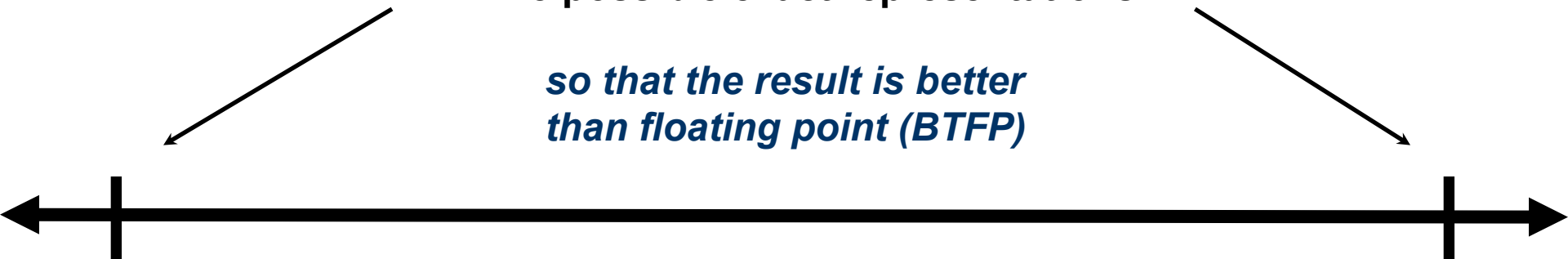


Non-exactly-representable values

round to one of the

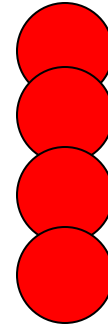
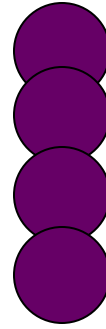
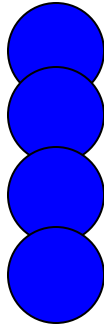
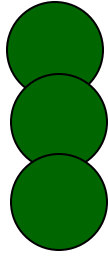
Two possible exact representations

*so that the result is better
than floating point (BTFP)*

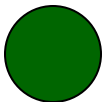


Restricted Faithful

p = .25

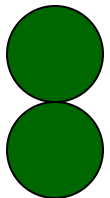
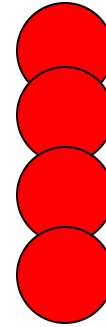
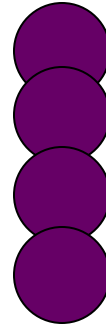
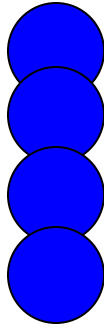
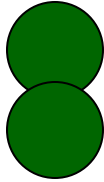


Values close to the left
always round there
(to the nearest)



Restricted Faithful

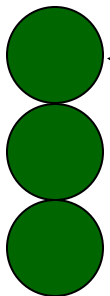
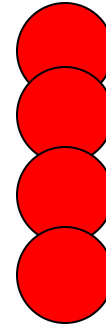
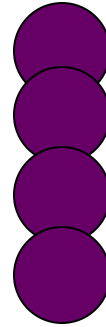
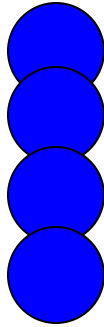
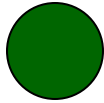
p = .25



Values close to the left
always round there
(to the nearest)

Restricted Faithful

p = .25

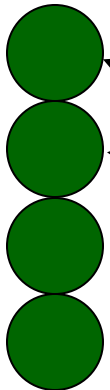
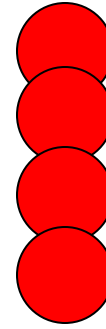
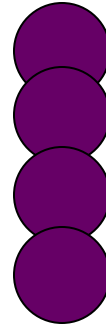
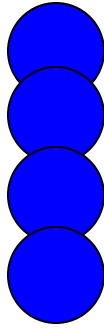


Values close to the left
always round there
(to the nearest)



Restricted Faithful

p = .25

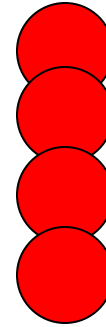
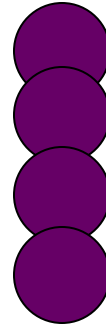
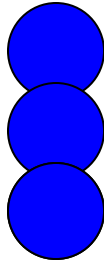


**Values close to the left
always round there
(to the nearest)**

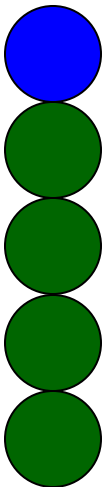


Restricted Faithful

p = .25

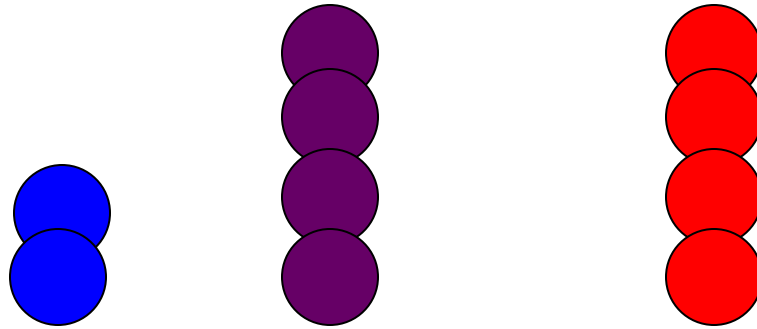


Values slightly left
of the midpoint
(blue) generally
round here

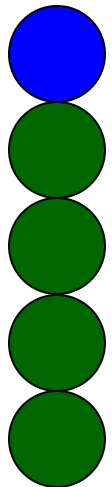


Restricted Faithful

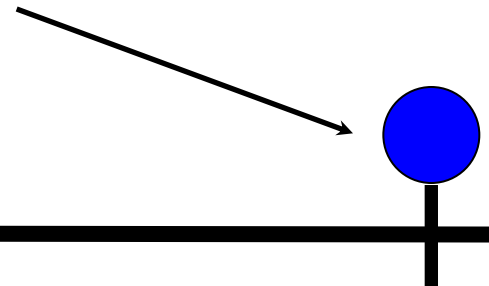
$p = .25$



Values slightly left
of the midpoint
(blue) generally
round here

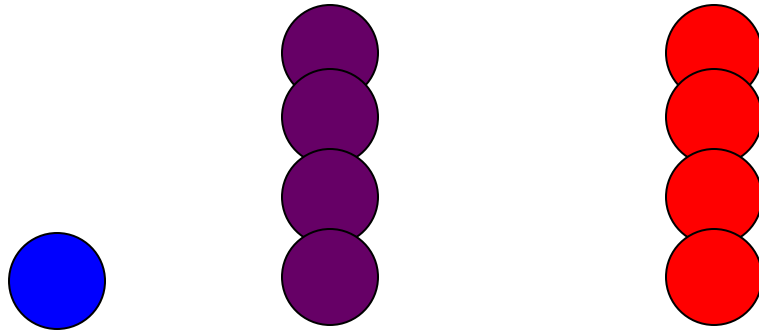


More than 1/4 of the blue
points round here
(adjusted for the width of
the midpoint region, δ)

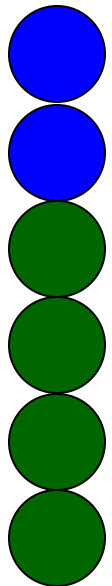


Restricted Faithful

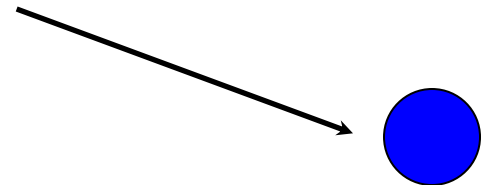
$p = .25$



Values slightly left
of the midpoint
(blue) generally
round here

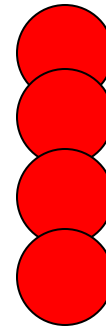
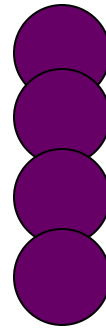


More than 1/4 of the blue
points round here
(adjusted for the width of
the midpoint region, δ)

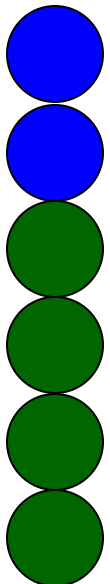


Restricted Faithful

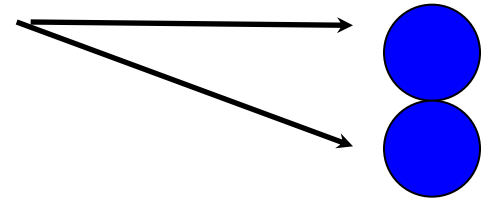
p = .25



Values slightly left
of the midpoint
(blue) generally
round here

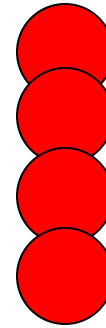
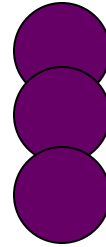


More than 1/4 of the blue
points round here
(adjusted for the width of
the midpoint region, δ)

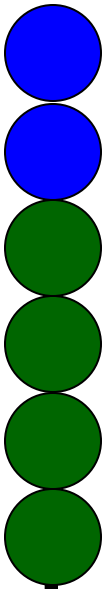
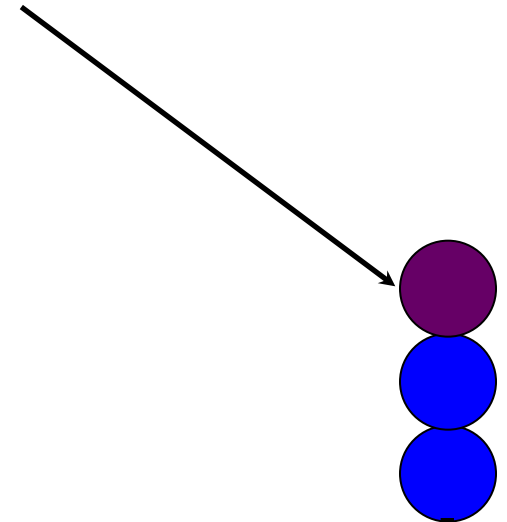


Restricted Faithful

p = .25

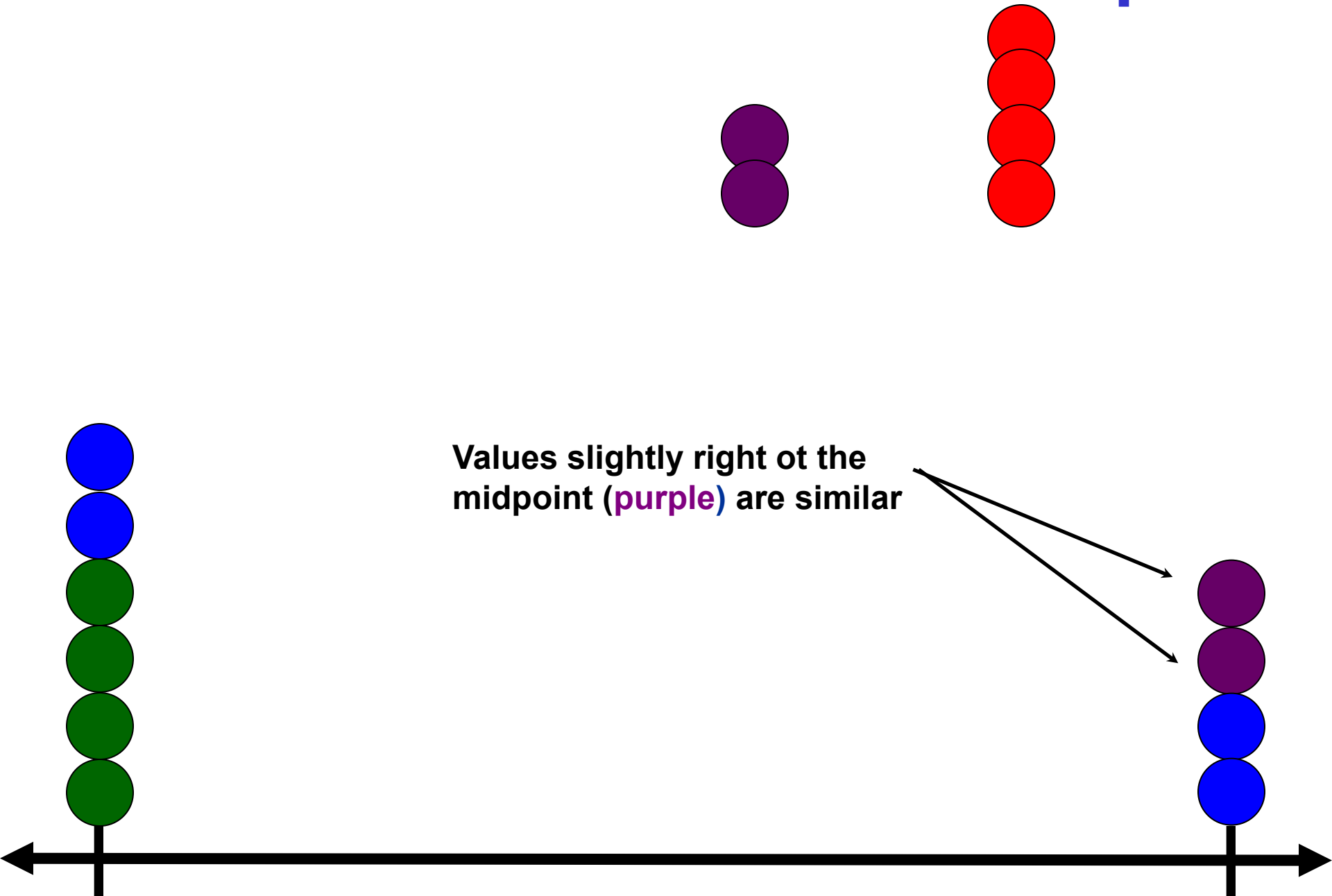


Values slightly right of the
midpoint (**purple**) are similar



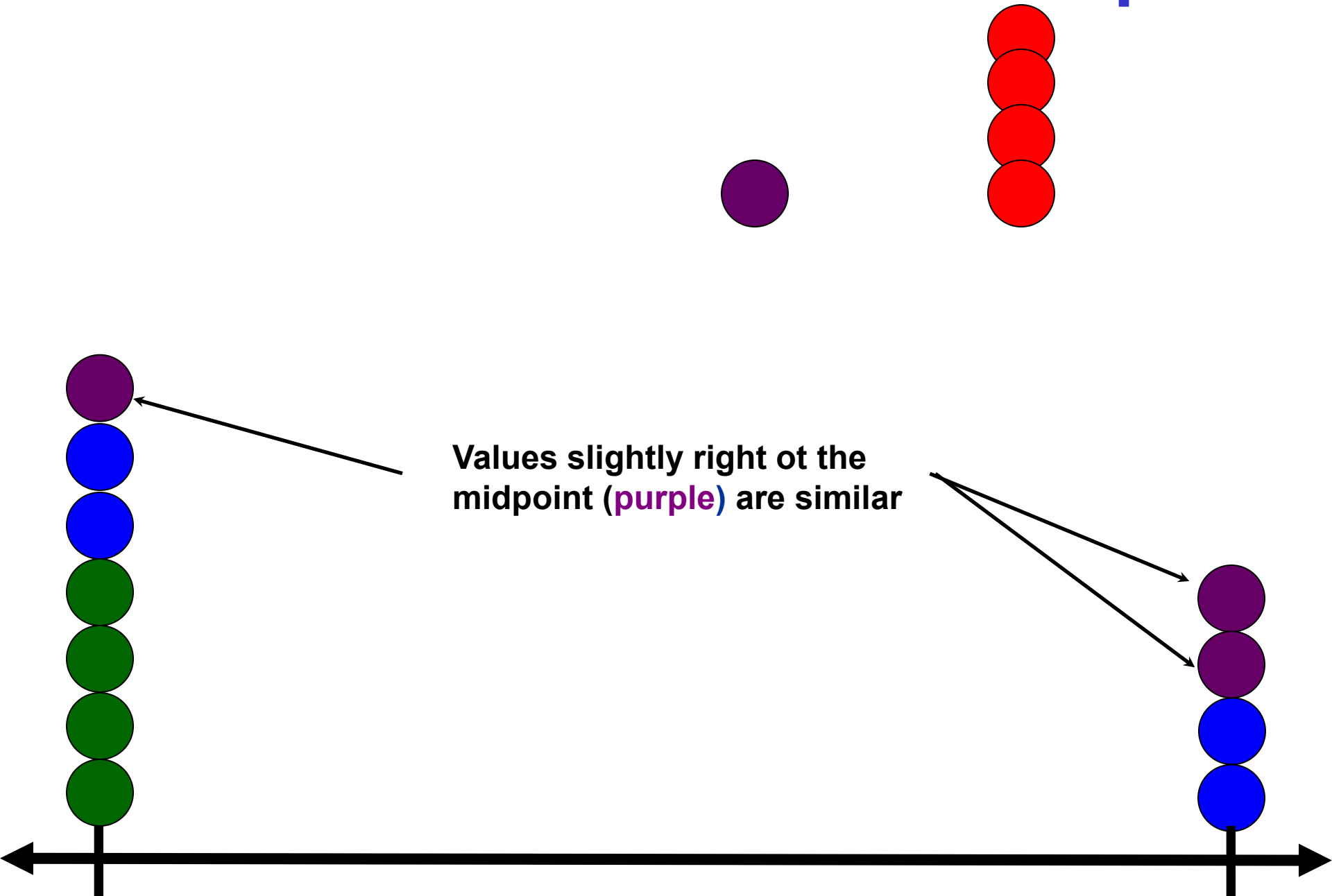
Restricted Faithful

$p = .25$



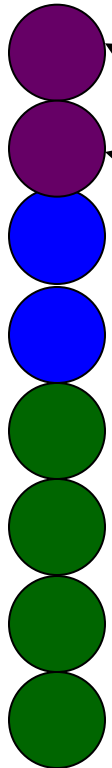
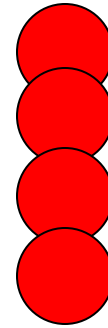
Restricted Faithful

$p = .25$

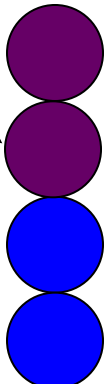


Restricted Faithful

$p = .25$

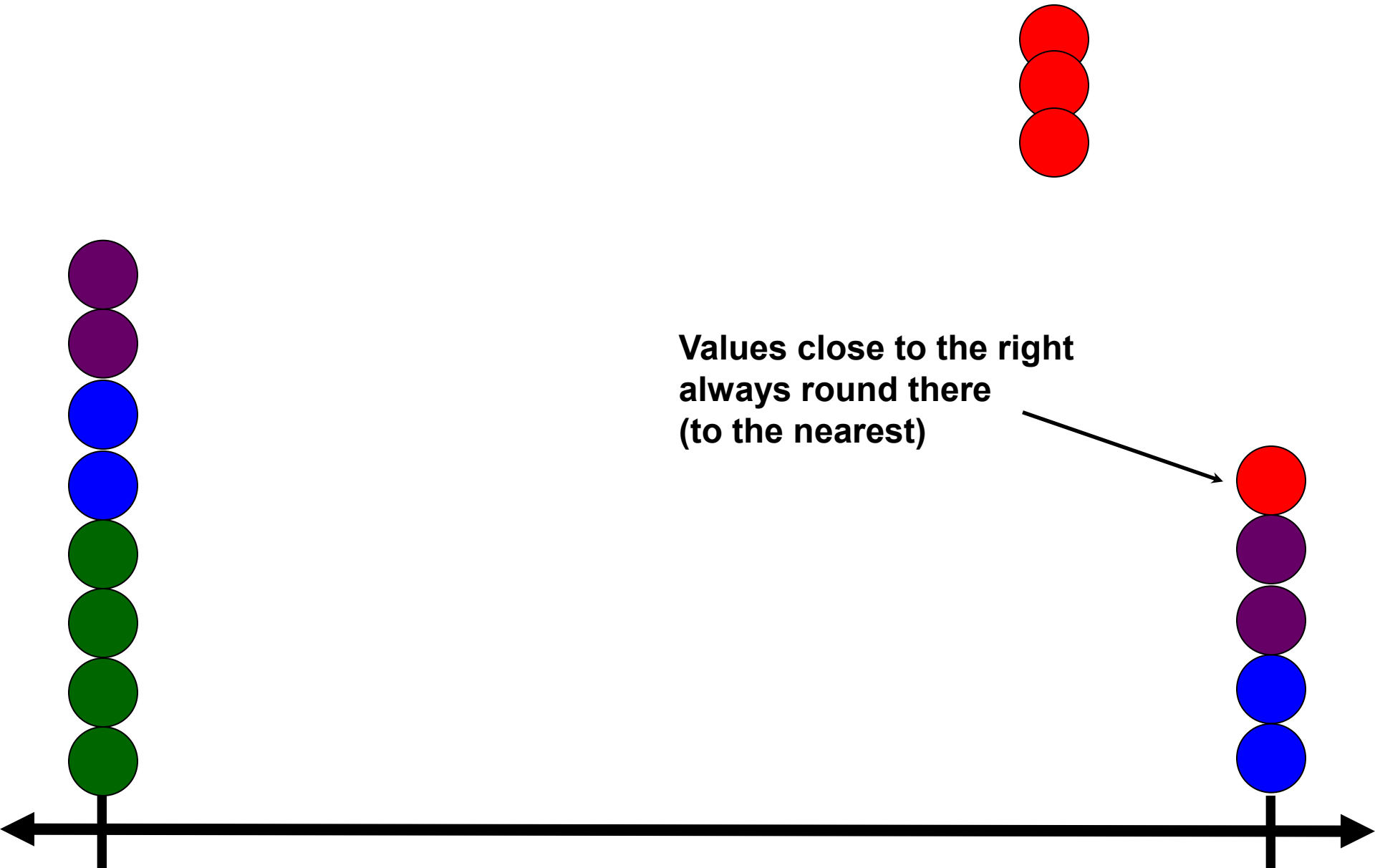


Values slightly right of the midpoint (**purple**) are similar



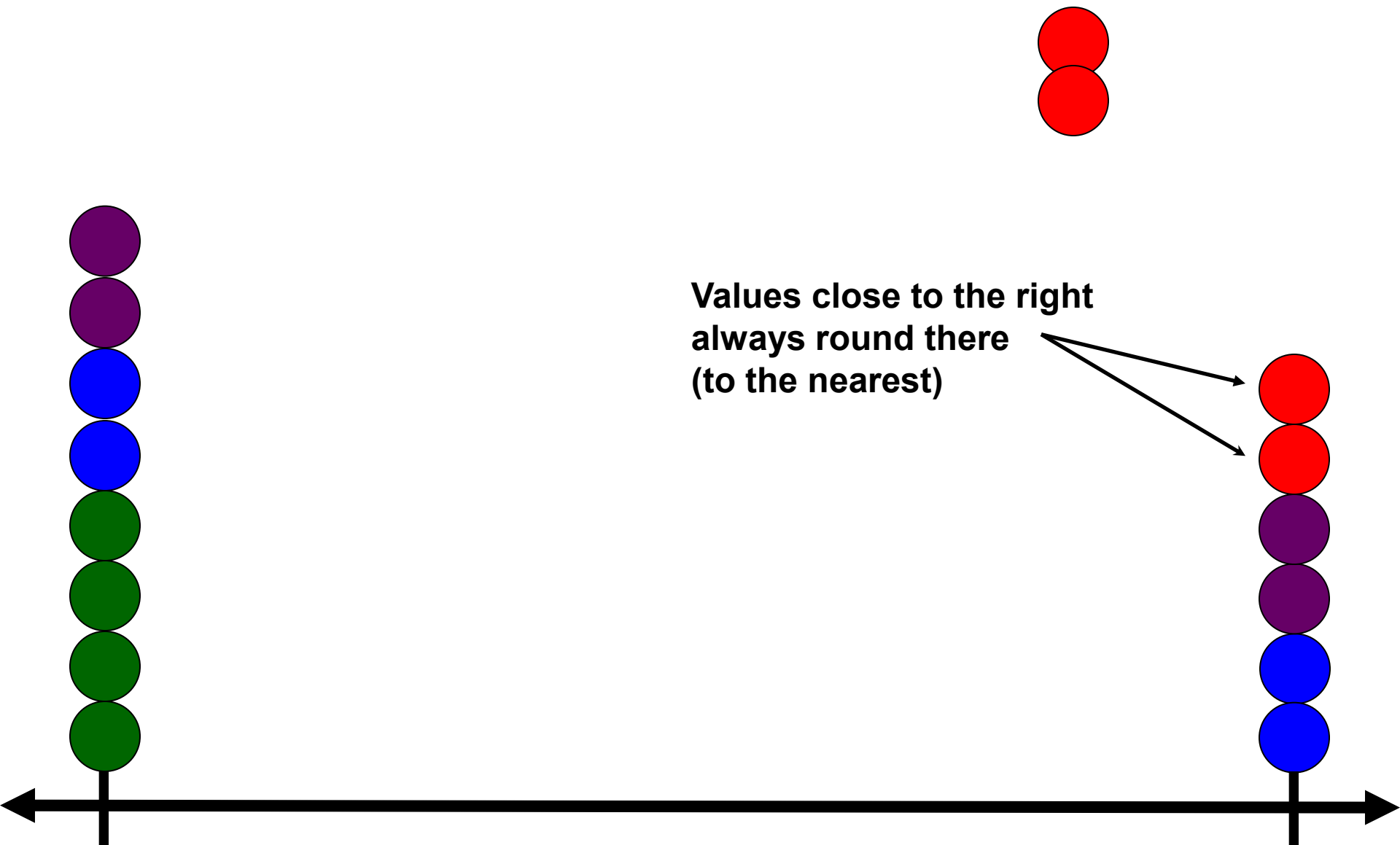
Restricted Faithful

p = .25



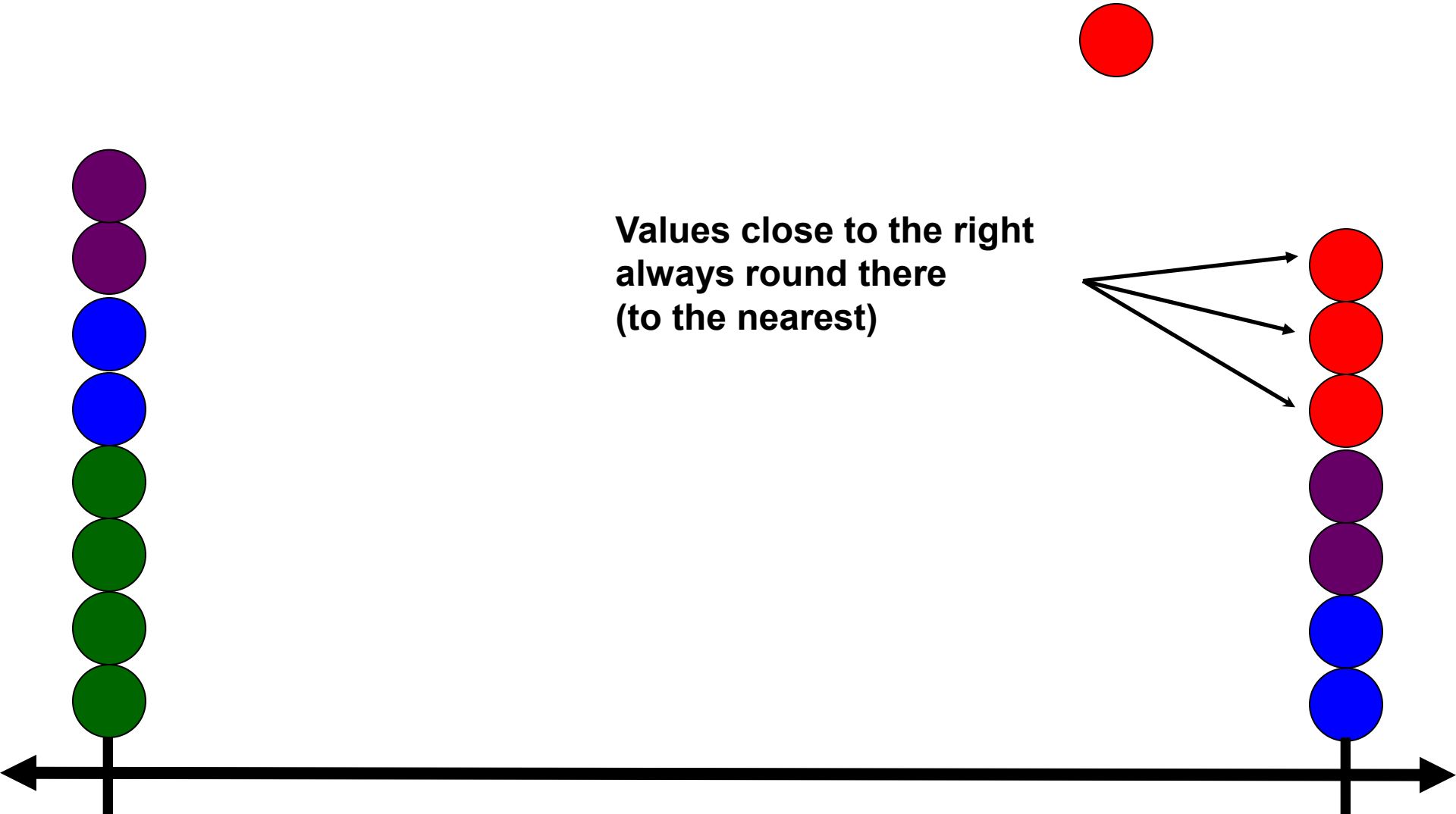
Restricted Faithful

$p = .25$



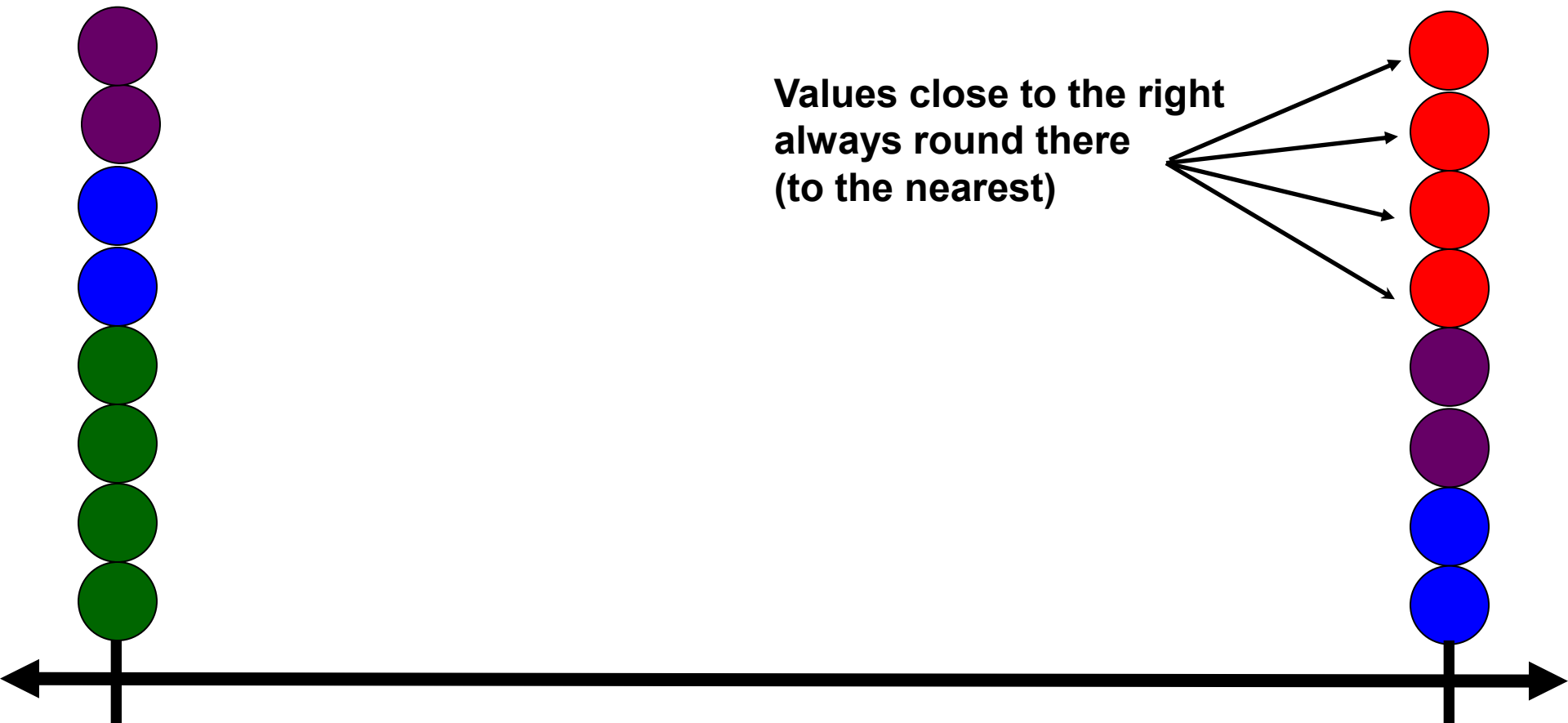
Restricted Faithful

$p = .25$



Restricted Faithful

$p = .25$



BTFP Figure 1: p. 246

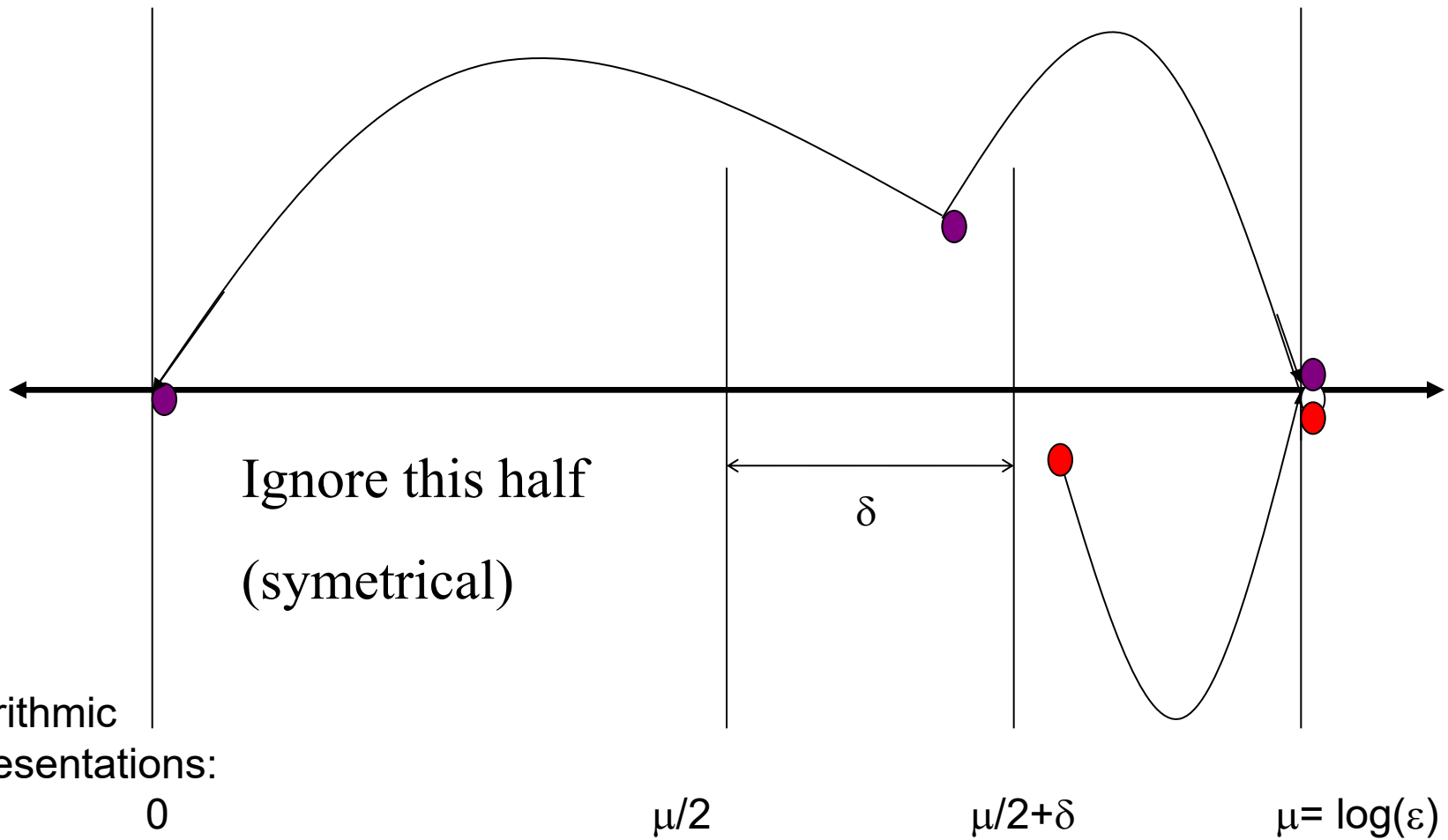
Real values:

1.0

$$\sqrt{\varepsilon} = 2^{\mu/2} \\ \cong 1 + \log_e(2)\mu/2$$

$$2^{\mu/2+\delta} \\ \cong 1 + \log_e(2)(\mu/2+\delta)$$

ε



Simulation

2ⁿ-point radix-two FFT

n 2ⁿ Complex MACs

4n 2ⁿ LNS additions and subtractions

error in first stage can propagate to all results

Compare F-bit arithmetics against 64-bit Floating Point

Input: n-point 25% duty square wave+white noise

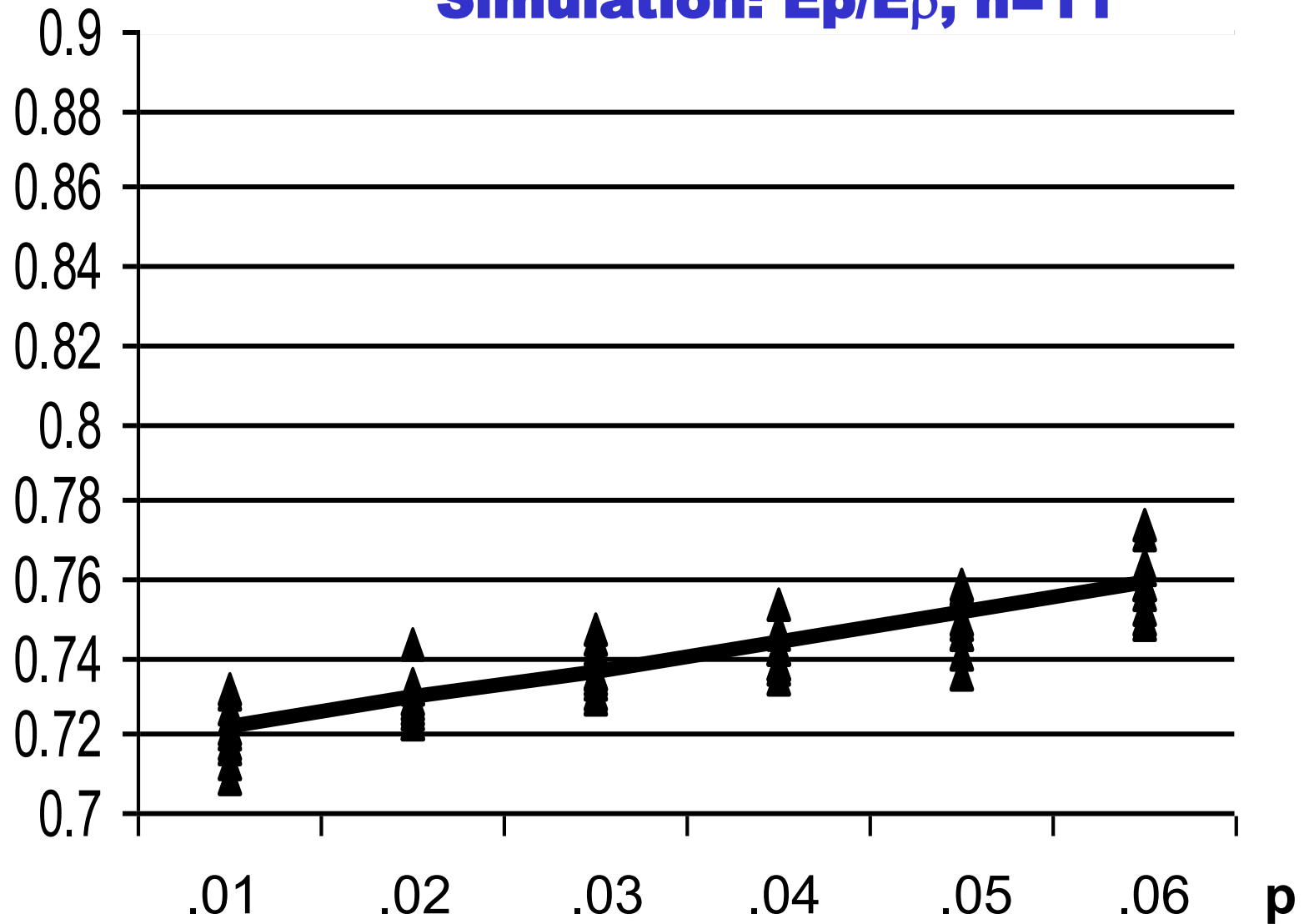
For $5 \leq n \leq 10$, $17 \leq F \leq 26$, note RMS errors:

E_p = RMS error for Restricted rounding

U_p = RMS error for Unrestricted rounding

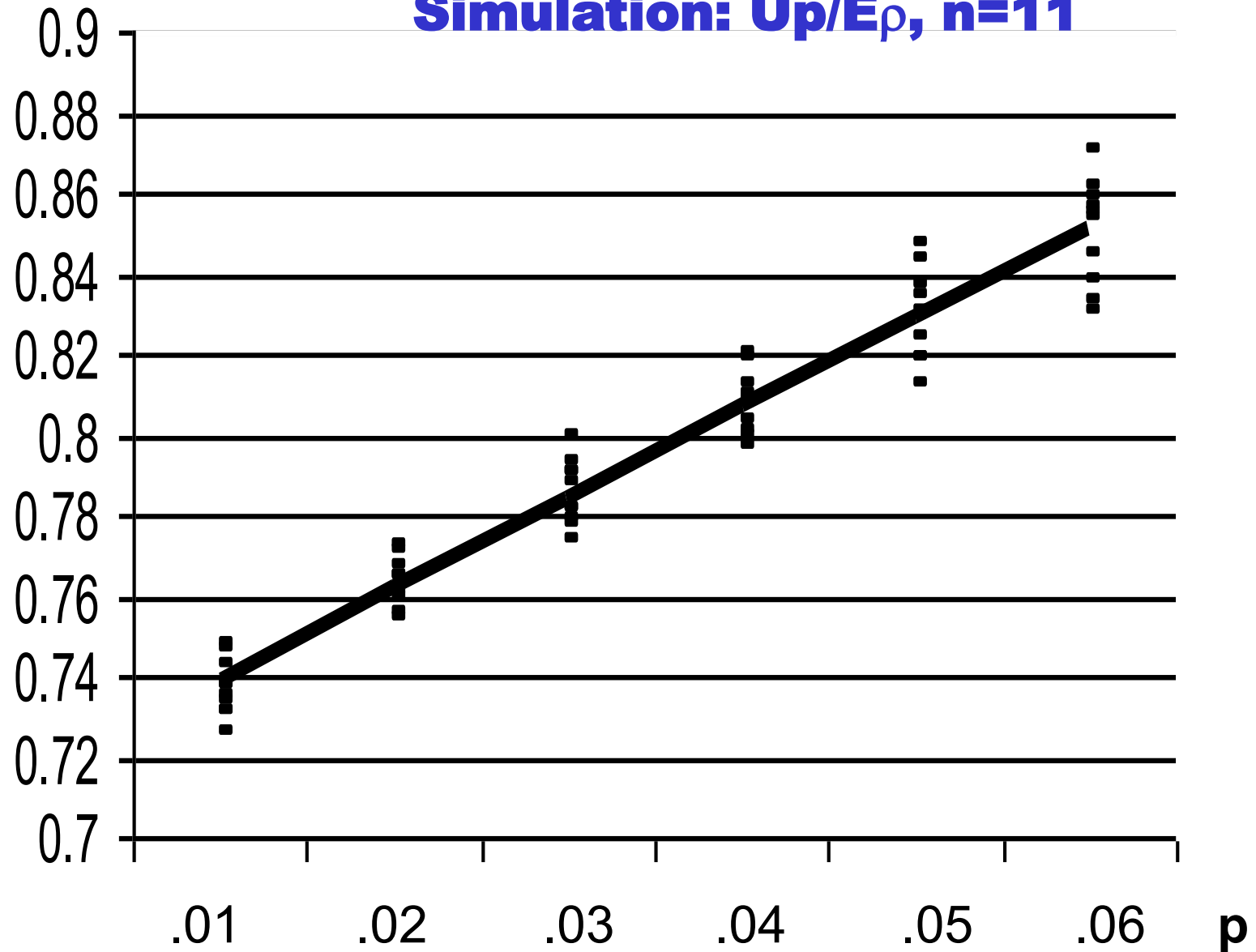
Re-run 250 times

Simulation: E_p/E_ρ , $n=11$



$$E_p/E_\rho \approx 0.71 + 0.73 p$$

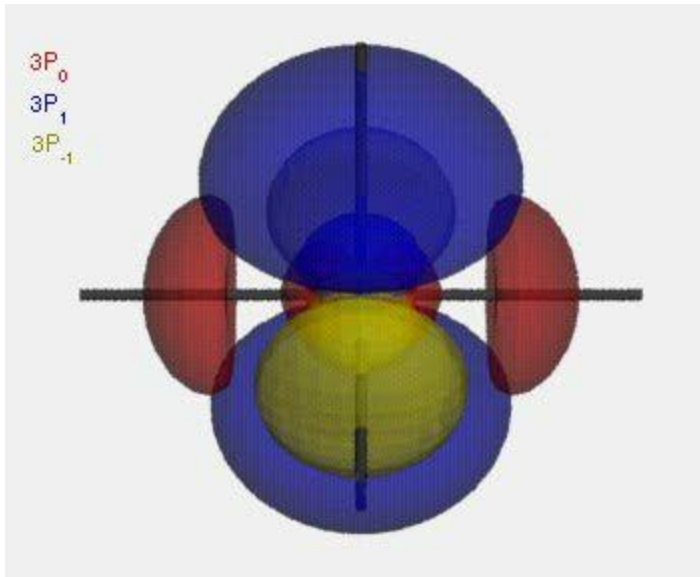
Simulation: $Up/E\rho$, $n=11$



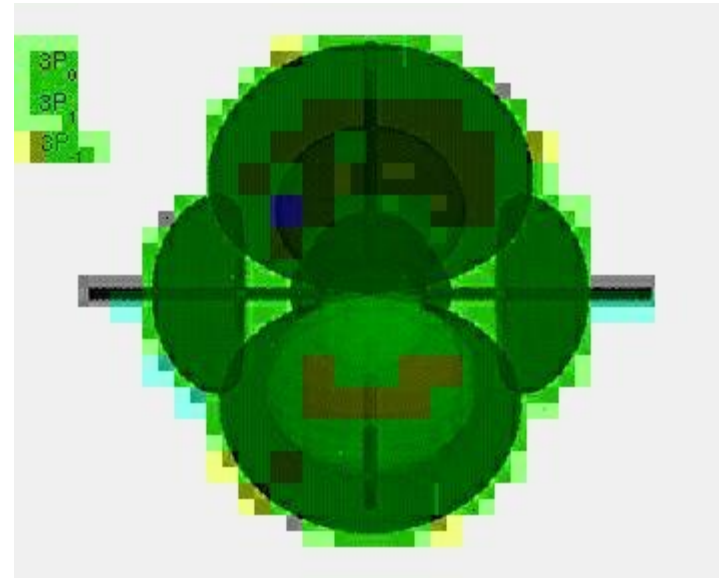
$Up/E\rho \approx 0.72 + 2.25 \rho$

MPEG Frame: Conventional Arithmetics

8 x 8 Inverse Discrete Cosine Transforms



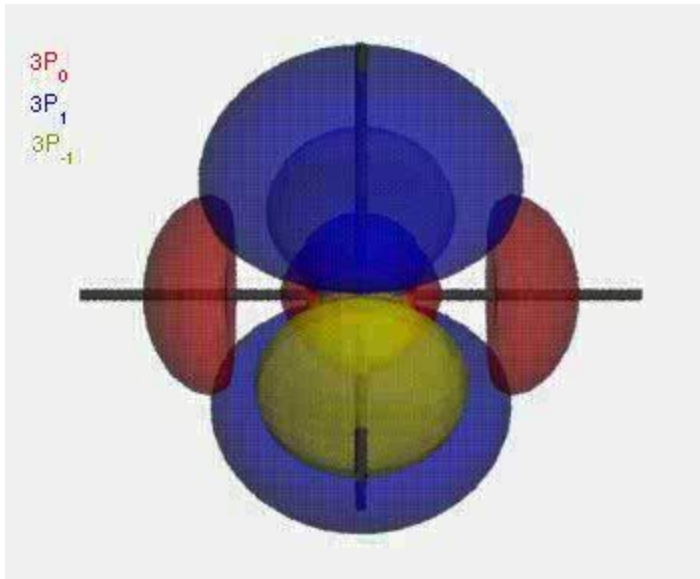
64-bit Floating Point
Round to Nearest (p=0)



14-bit Fixed Point
Round to Nearest

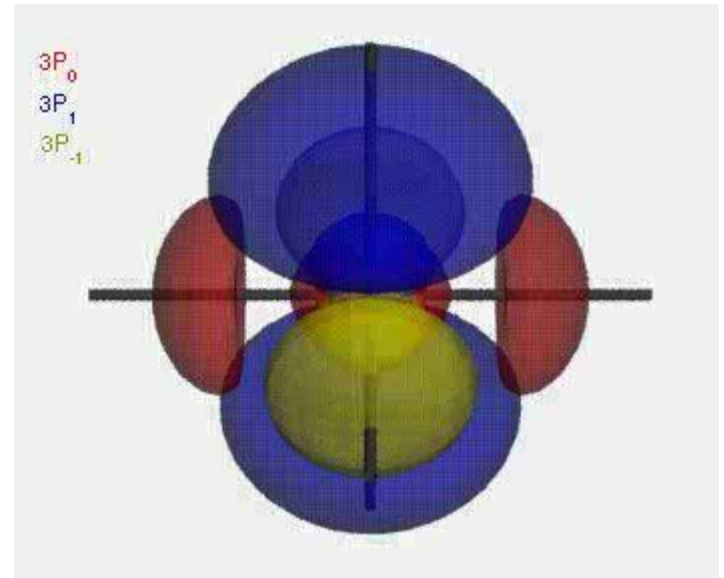
MPEG Frame: LNS Arithmetic

8 x 8 Inverse Discrete Cosine Transforms



11-bit LNS

Round to Nearest (p=0)

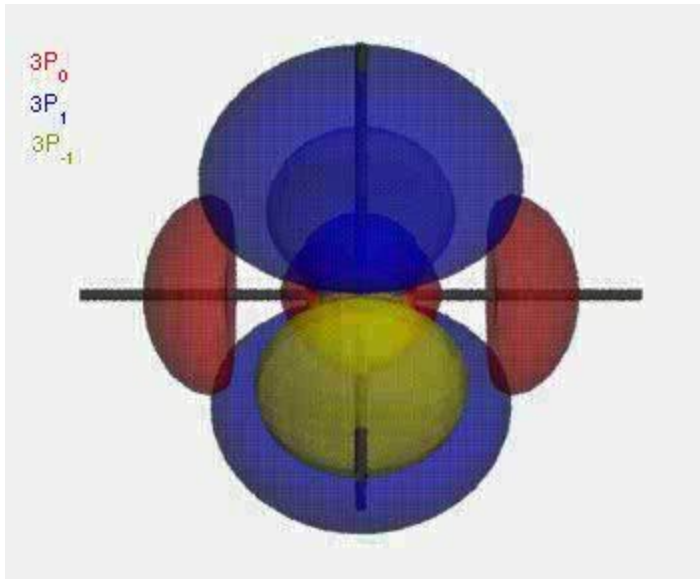


11-bit LNS

Unrestricted Faithful (p=.5)

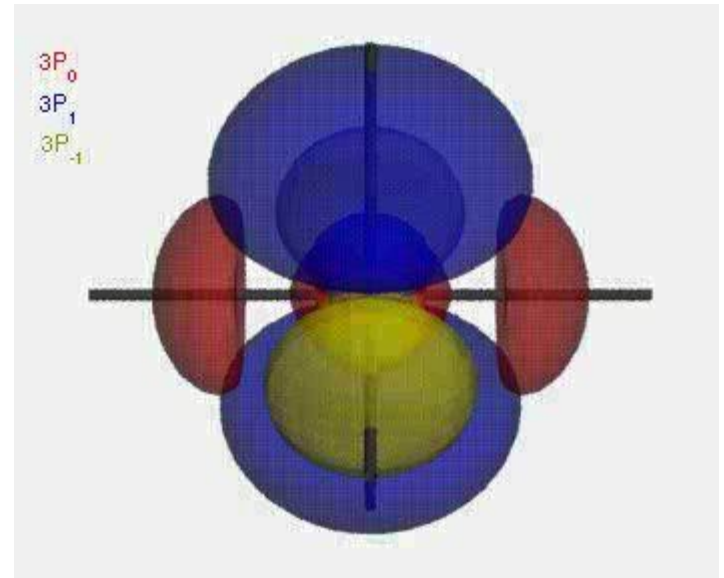
MPEG Frame: LNS Arithmetic

8 x 8 Inverse Discrete Cosine Transforms



10-bit LNS

Round to Nearest ($p=0$)



10-bit LNS

Unrestricted Faithful ($p=.5$)

Conclusions

Unrestricted faithful LNS rounding is good enough for:

low-precision FFT

MPEG-1 decoding

similar multimedia and DSP applications...

Reduces table size 3- to 5- fold for LNS equivalent to 32-bit FP

Expect an order of magnitude improvement for MPEG-1

